Representation constraints and polynomial-depth learning in Gaussian Bayesian networks

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Abstract

We study Gaussian-prior tanh networks conditioned on a clean output margin throughout the Boolean cube and on noiseless training signs. For every fixed $\zeta > 0$ and exponent $\gamma \geq 1 + \zeta$, a posterior trained on n examples at depth $L \geq 3$ and sufficiently large polynomial width assigns mass at least $1 - \beta$ to Boolean functions with same-depth unit-margin representatives whose product of Frobenius norms is polynomial in d, n. A general polynomial sample theorem would therefore require a corresponding approximation representation theorem.

Conversely, a target with depth D_0 and product norm P has polynomial sample complexity, simultaneously at every width above its actual bounded-weight network complexity, after only $O(\log(d+D_0+P))$ additional learner layers. The original target width and coefficient precision do not enter. We prove a norm-preserving width compression theorem and a further global-margin normal form whose entire Boolean description is polynomial in P, independently of original depth and width, apart from specifying at most P^2 relevant coordinate indices. At sufficiently large depth and width, the minimum Gaussian energy above its depth baseline is comparable to the logarithm of this intrinsic product complexity, together with the input-normalization cost.

The positive results include every Boolean halfspace, independently of its original geometric margin and coefficient precision, at every learner depth $L \geq 2$. We also prove polynomial sample bounds for bounded-weight teachers with at most K possible hidden states after a fixed number of layers. The bound is polynomial in K, teacher resources, and learner depth; no hidden-state separation is assumed. For fan-in-two Boolean circuits of size s and depth h, put $S = \max\{16, 2 + d + s\}$. At every learner depth $L \geq \max\{2, h\}$, we obtain the upper bound $S^{O_\zeta(1+h/\log\log S)}$ times a polynomial in learner depth and statistical parameters. Apart from the cited Goldmann–Karpinski threshold-to-majority simulation, the Hesse–Allender–Barrington constant-depth threshold-circuit arithmetic, and Gaussian isoperimetry, we give all analytic and representation arguments needed for these results. A sample bound jointly polynomial in width and depth for arbitrary small deep circuits remains unresolved. A separate critical-prior paper treats width-first learning at $\gamma = 1$.

1 Model and main conclusions

The target $f : \{-1, 1\}^d \to \{-1, 1\}$ is learned from independent examples $(x_i, f(x_i))$ drawn from an arbitrary input distribution D. A learner of depth $L \geq 2$ has $L - 1$ hidden layers, each of width N, and computes

$$h^0 = (1, x), \qquad h^s = \tanh(W^s h^{s-1}) \quad (1 \leq s < L), \qquad z_w(x) = v^T h^{L-1}.$$

There are no biases after the augmented input. Its Gaussian energy is

$$\kappa_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{L-1} \|W^s\|_F^2 + N\|v\|_2^2.$$

The raw Gaussian has independent entry variances $t^2/d, t^2/N, t^2/N^2$, respectively. Put $\sigma^2 = t^2/N$; the exponent-γ convention is $\sigma^2 = N^{-\gamma}$. In energy coordinates its density is proportional to $\exp[-\kappa_N/(2\sigma^2)]$.

The posterior Q_S is the raw Gaussian conditioned on

$$|z_w(x)| > 1/2 \quad \text{for every } x \in \{-1,1\}^d, \qquad f(x_i)z_w(x_i) > 0 \quad \text{for every training input.}$$

"Unrestricted" retains this all-input output-margin condition and means that no additional internal robustness restriction is imposed. Let $C_L(f)$ be the least width admitting a depth-L representation with all physical weights bounded by that width and signed output greater than $1/2$ everywhere. This is the actual bounded-weight network complexity at the specified learner depth.

Our new representation constraint is Theorem 3.2. At every learner depth $L \geq 3$, for every consistently labeled sample of size at most n, every $0 < \beta < 1$, and all widths above a polynomial threshold, posterior mass at least $1 - \beta$ is supported on Boolean functions admitting a unit-margin depth-L representative with product of Frobenius norms polynomial in d, n. The threshold is polynomial in $d, n, L, \log(2/\beta)$, with degree depending on the fixed gap $\zeta > 0$ above exponent one. If L is at least a constant times $\log(d+n)$, one additional layer gives a representative of energy $L + O(\log(d+n))$. Corollary 3.3 turns any function-level approximation-energy lower bound into an actual posterior error lower bound. We do not establish such a barrier for an explicit small deep target here.

Our new positive conclusion, Theorem 4.3, covers every Boolean halfspace with sample size polynomial in d, L, ε^{-1} and $\log(2/\delta)$, simultaneously for every $N \geq C_L(f)$ and every $0 < \sigma^2 \leq N^{-(1+\zeta)}$. Unlike a direct affine implementation, the alternative shallow representation does not depend on the target's geometric margin or original weight magnitudes. For halfspaces, the only external representation ingredient is the classical Goldmann–Karpinski threshold-to-majority simulation; the remaining analytic arguments are included below.

Theorem 7.3 supplies a different positive class: a bounded-weight teacher of depth D whose state after a fixed number of initial layers has at most K values. Its sample bound is polynomial in K, teacher resources, and learner depth, at every $L \geq D$ and every actual width $N \geq C_L(f)$. Only the teacher has a finite-state bottleneck; the prior receives no additional restriction. For a bottleneck immediately after the first layer, the bound at every learner depth $L \geq 6$ is even independent of the remaining computation. Examples include first layers driven by a fixed number of groupwise bit counts, or fixed-width first layers with polynomially bounded integer weights. An arbitrary growing prefix depth or exponential state count is not covered by this polynomial assertion.

The positive and negative directions meet at a function-level representation question. Theorems 10.1 and 13.3 give, respectively, a partial converse after logarithmic added depth and a finite Boolean description controlled by product norm alone. Theorem 15.1 then relates intrinsic product complexity to minimum excess Gaussian energy at sufficiently large depth and width. None of these statements identifies ordinary circuit size with product complexity. For a general fan-in-two circuit of size s and depth h, put $S = \max\{16, 2 + d + s\}$. At every $L \geq \max\{2, h\}$, the block construction gives $S^{O_\zeta(1+h/\log\log S)}$ sample dependence, improving the direct $S^{O_\zeta(h)}$ bound while retaining a gap from a joint polynomial. Every fixed-depth threshold-circuit family is covered by a polynomial bound; thus an explicit superpolynomial-sample counterexample within NC^1, at growing learner depth, would separate NC^1 from TC^0.

Section 16 clarifies one possible route from distribution-wise approximation to an exact representation. Arbitrary added depth can be supplied at constant product cost. Moreover, weak classifiers sharing one nonlinear feature map can be combined at polynomial product cost. A general closure theorem for classifiers with different feature computations remains open; simply placing

their networks in parallel can incur an exponential depth factor. This separates an architecture-level obstruction from the unresolved function-level representation question.

We measure error primarily by posterior Gibbs error $\mathbb{E}_{w\sim Q_S} \Pr_D\{\operatorname{sign} z_w(X) \neq f(X)\}$. If $p_S(x)$ is posterior probability of the correct label, then

$$D\{x : p_S(x) \leq q\} \leq \frac{\mathrm{GibbsErr}(Q_S)}{1-q} \qquad (0 < q < 1).$$

Thus all positive results also cover correct-answer probability thresholds $2/3$ and $1/2$, after constant changes in the Gibbs error target. Posterior mixtures require no special assumption. The lower bound in Corollary 3.3 states its separate consequence for the $2/3$ criterion explicitly.

The polynomial degrees may depend on ζ; no uniform limit as $\zeta \downarrow 0$ is claimed. All results concern statistical sample complexity, not efficient posterior computation. The draft has received independent AI-agent proof audits, but is not Lean-certified.

2 Changing the output margin at constant additional energy

Lemma 2.1 (One-layer margin transfer)**.** *Suppose a width-N, depth-L tanh network with linear readout computes a Boolean function g with global signed margin at least $r > 0$ and Gaussian energy K. For every $r' > 0$, there is a network at the same width and depth $L+1$ computing g with signed margin at least r' and energy at most*

$$K + (r'/r)^2 + (r')^2.$$

The construction uses no biases after the original first layer.

Proof. Let v be the old linear readout, and write $c^2 = N\|v\|_2^2$. The global margin implies $c \geq r > 0$, because the hidden coordinates have absolute value at most one. Define

$$a^2 = \frac{r'}{cr}, \qquad b = \frac{r'}{\tanh(ar)}.$$

Replace the original readout by a hidden layer all of whose rows are av^T, and give the new readout the constant entries b/N. The new output is exactly $b\tanh(az(x))$. Oddness and monotonicity give the claimed signed margin. The new two-layer energy is $a^2c^2 + b^2$ in place of the old readout energy c^2.

For every real u, $|\sinh u| \geq |u|$, so

$$\tanh^2 u = \frac{\sinh^2 u}{1+\sinh^2 u} \geq \frac{u^2}{1+u^2}.$$

Consequently

$$b^2 \leq (r')^2 + \frac{(r')^2}{a^2r^2}, \qquad a^2c^2 + b^2 \leq 2c\frac{r'}{r} + (r')^2.$$

Completing the square gives $-c^2 + 2cr'/r \leq (r'/r)^2$, proving the result. □

For example, a globally clean residual network with signed margin at least $3/8$ can be converted to unit margin by adding one layer and at most $73/9$ energy. The increase is independent of the old readout cost; simply multiplying the old readout need not have that property.

Define the approximation energy

$$\mathcal{E}_{L,r}(f;\epsilon) = \inf\left\{\kappa_N(W): \begin{array}{l} N \geq 1, \quad |z_W(x)| \geq r \text{ for all Boolean } x, \\ \Pr_{X\sim D}\{\operatorname{sign} z_W(X) \neq f(X)\} < \epsilon \end{array}\right\},$$

where D is a specified input distribution and an empty infimum is $+\infty$. The lemma implies the exact comparison

$$\mathcal{E}_{L+1,r'}(f;\epsilon) \leq \mathcal{E}_{L,r}(f;\epsilon) + (r'/r)^2 + (r')^2.$$

Thus a unit-margin approximation-energy lower bound at one extra layer implies a lower bound at any fixed smaller positive margin, losing only a universal additive constant when the two margins are fixed.

3 A posterior energy ceiling and a necessary representation condition

The following argument identifies a necessary condition for the desired jointly polynomial unrestricted learning theorem. It concerns the actual finite Gaussian scales above exponent one, not just a cold limit. Throughout, the prior is conditioned on the original global clean margin $|z(x)| > 1/2$ and on the observed training signs.

Lemma 3.1 (A uniform memorizing witness)**.** *For integers $d, n \geq 1$, there are explicitly polynomial quantities $M = M(d,n)$ and $T = T(d,n) \geq 1$ such that every consistently labeled sample of at most n Boolean inputs admits a globally unit-margin interpolating network at every width $N \geq M$ and every depth $L \geq 3$, with energy at most*

$$K = 2 + (L+14)(2T)^{1/(L-1)}.$$

One may take

$$C = n+1, \quad R = \max\{4(n+1), (d+1)\log(16(n+1))\},$$
$$M = \left\lceil \max\{4, C, 2R, dC(d+1)R^2\} \right\rceil, \qquad T = MCR^2.$$

In particular, if $L - 1 \geq \log(2T)$, then

$$K \leq L + B(d,n), \qquad B(d,n) = 16 + 32\log(2T) = O(\log(d+n)).$$

Proof. Let $p_1, \ldots, p_m$ be the distinct negatively labeled sample points, where $0 \leq m \leq n$. Set $a = \log(16(n+1))$ and

$$q_j(x) = \frac{d - p_j \cdot x}{2}, \qquad \tau_j(x) = \tanh\big(a(1/2 - q_j(x))\big).$$

Each q_j is a nonnegative integer on the cube and is zero exactly at p_j. Thus τ_j differs by at most $2e^{-a}$ from the sign which equals $+1$ at p_j and -1 elsewhere. Since the p_j are distinct,

$$z(x) = -2\left[m - 1 + \sum_{j=1}^{m} \tau_j(x)\right]$$

has the desired interpolating signs, equals $+2$ or -2 in the ideal sign-feature limit, and differs from those values by at most $4me^{-a} < 1/4$. The formula also gives the constant $+2$ when $m = 0$.

Implement its constant term using a first-layer constant feature $1/2$, whose input bias is $\operatorname{atanh}(1/2)$. Pad with zero neurons to $C = n + 1$. Every physical weight is bounded by the stated R, and the global signed margin is greater than $7/4$.

Zero-padding this shallow network to M makes its first-layer energy at most one and its weighted readout energy at most T. Balanced cloning to any $N \geq M$ gives first energy at most two and readout energy at most $2T$. The scalar-transmission construction of Lemma 19.3, applied to this one-hidden-layer witness, gives the displayed bound at every $L \geq 3$.

For the final assertion, put $u = \log(2T)/(L-1) \leq 1$. The inequality $e^u \leq 1 + 2u$ on $[0, 1]$, and $(L + 14)/(L - 1) \leq 16$, give

$$2 + (L + 14)e^u \leq L + 16 + 32\log(2T).$$

□

Theorem 3.2 (A finite-scale posterior energy ceiling)**.** *Fix $d, n \geq 1$, $L \geq 3$, $\zeta > 0$, and $0 < \beta < 1$. Use the quantities M, T, K above and put*

$$Q_* = ((K + 2)/L)^{L/2}.$$

There is a width threshold polynomial in $d, n, L, \log(2/\beta)$, with degree depending only on ζ, such that at every width above this threshold and every $0 < \sigma^2 \leq N^{-(1+\zeta)}$, the following holds for every *consistently labeled sample of size at most n.*

With posterior probability at least $1 - \beta$, the posterior Boolean function has a same-width, same-depth realization with global signed margin greater than $3/8$ and energy at most $K + 1$. Rescaling its readout gives a unit-margin realization at the same width and depth, with product of layer Frobenius norms bounded by

$$P_F^2 \leq d^{-1}e^{19}(2T)^{3/2}.$$

Alternatively, it has a same-width realization at depth $L + 1$, with unit margin and energy at most $K + 82/9$.

When $L - 1 \geq \log(2T)$, the last energy bound is $L + O(\log(d + n))$.

Proof. Use Gaussian energy coordinates $A_1 = \sqrt{d/N}W^1$, $A_s = W^s$ internally, and $A_L = \sqrt{N}v$. Their entries are independent $\mathcal{N}(0, \sigma^2)$ and their squared Euclidean norm is κ_N. Write $t^2 = N\sigma^2$.

We recall the explicit estimates from the Gaussian-distance bridge. There is a closed, convex, centrally symmetric Gaussian set $\mathcal{T}$ of probability at least $1/2$ whose normalized block operator norms are at most tb, where

$$b^2 = \max\{2(d + 1)\log 2 + 4\log 8, 16\log 9 + 8\log(8L)\}.$$

Put

$$A = \left(\frac{K + 2}{L - 1}\right)^{(L-1)/2}, \qquad \xi = \min\left\{1, \frac{1}{4\sqrt{L(K + 1)}}, \frac{1}{8\sqrt{2}LA}\right\}.$$

If $tb \leq \xi$, adding any $B \in \mathcal{T}$ to any network of energy at most $K + 1$ changes its output uniformly on the cube by at most $1/8$. This follows by telescoping the layer perturbations and applying arithmetic–geometric mean to the other $L - 1$ layer norms.

For the realized sample, choose the unit-margin memorizing witness w_S above. Its shifted set $w_S + \mathcal{T}$ lies in the conditioning event. Central symmetry and the Gaussian shift identity give

$$\mu(w_S + \mathcal{T}) \geq \tfrac{1}{2}e^{-K/(2\sigma^2)}.$$

Gaussian isoperimetry gives $\mu\{\mathrm{dist}_\kappa(w,\mathcal{T})^2 > K+1\} \le e^{-(K+1)/(2\sigma^2)}$. Dividing yields

$$Q_S\{\mathrm{dist}_\kappa(w,\mathcal{T})^2 > K+1\} \le 2e^{-1/(2\sigma^2)}.$$

On the complementary event, subtract the metric projection of w onto $\mathcal{T}$. The residual U has energy at most $K+1$, and its output differs by at most $1/8$ everywhere. The original clean margin is greater than $1/2$, so the residual has the same Boolean function and margin greater than $3/8$. Multiplying its readout by $8/3$ gives unit margin without changing its depth. Lemma 2.1 adds at most $73/9$ energy to obtain unit margin at one extra layer.

It suffices to take

$$N \ge \left\lceil \max\left\{M, 2, (b/\xi)^{2/\zeta}, 2\log(2/\beta)\right\}\right\rceil.$$

Indeed, $t \le N^{-\zeta/2}$ and $\sigma^{-2} \ge N$, so both estimates hold simultaneously for every permitted scale. Since $A \le \sqrt{e}Q_*$, this is a polynomial threshold in those quantities. In fact, put $q^2 = (2T)^{1/(L-1)} \ge 1$. Since $K+2 \le (L+18)q^2$, one has for every $L \ge 3$

$$Q_*^2 \le e^{18}(2T)^{L/(L-1)} \le e^{18}(2T)^{3/2}.$$

Thus the threshold is polynomial in $d, n, L, \log(2/\beta)$ for all the stated depths. Since $K+1 \le (L+17)q^2$, the same-depth residual and arithmetic–geometric mean give, after rescaling the readout,

$$dP_F^2 \le \frac{64}{9}\left(\frac{K+1}{L}\right)^L \le \frac{64}{9}e^{17}(2T)^{L/(L-1)} \le e^{19}(2T)^{3/2}.$$

The additive-excess conclusion follows from the memorizing-witness lemma. All constants and the set $\mathcal{T}$ are uniform over the realized sample; only the translating witness depends on its labels. □

Corollary 3.3 (Approximation-energy barriers force posterior error)**.** *Let f and its input distribution D be arbitrary. Under the conditions of Theorem 3.2, suppose*

$$\mathcal{E}_{L+1,1}(f;\epsilon) > K + 82/9.$$

Then, for every realized sample of at most n examples labeled by f, the posterior Gibbs error is at least $(1-\beta)\epsilon$. The same conclusion holds if no globally unit-margin depth-L network with $P_F^2 \le d^{-1}e^{19}(2T)^{3/2}$ has D-error less than ϵ, allowing arbitrary finite widths. If $p_S(x)$ is the posterior probability of the correct label, then

$$\Pr_D\{p_S(X) \le 2/3\} \ge \max\left\{0, \frac{3(1-\beta)\epsilon - 1}{2}\right\}.$$

Conversely, if some sample gives posterior Gibbs error at most r, then for every $r' > r/(1-\beta)$,

$$\mathcal{E}_{L+1,1}(f;r') \le K + 82/9.$$

There is also a globally unit-margin approximation of error less than r' at the original depth L, with $P_F^2 \le d^{-1}e^{19}(2T)^{3/2}$.

Proof. The energy ceiling puts posterior mass at least $1-\beta$ on Boolean functions with unit-margin representatives of energy at most $K + 82/9$. Under the barrier, every such function has error at least ϵ. The same posterior event has the same-depth product-norm realization in the theorem, proving the alternative barrier statement. For the confidence claim, if the displayed confidence-error probability is e, then Gibbs error is at most $(1-e)/3 + e$. For the converse, condition the posterior on its event of mass at least $1-\beta$. Its average error is at most $r/(1-\beta)$, so for every larger r' it contains a function of error less than r'. □

Thus a jointly polynomial sample theorem for all small deep targets would imply a representation theorem at every $L \geq 3$: every target learned with polynomially many samples must admit a constant-error, unit-margin approximation at the same depth L whose product of layer Frobenius norms is polynomial in the target parameters and d. At depths sufficiently large compared with $\log(d+n)$, every such target must admit a constant-error approximation of energy $L + O(\log(d+n))$ at depth $L+1$. In particular, if n is polynomial in d, C, L, the required excess energy is only $O(\log(d+C+L))$. This conclusion does not depend on choosing the product-norm proof strategy for the upper bound; it is forced by the posterior itself. No example of a small target violating this necessary condition has been established here.

4 Boolean halfspaces need no geometric-margin assumption

The shallow-target theorem applies to every Boolean linear threshold function, even when its most natural affine representation has extremely small geometric margin. The relevant fact is a classical circuit simulation theorem; retaining the original affine representation would miss this conclusion.

Lemma 4.1 (Classical threshold-to-majority simulation)**.** *There is a universal polynomial p with the following property. Every Boolean threshold function $f : \{-1,1\}^d \to \{-1,1\}$ can be written*

$$f(x) = \operatorname{sign}\left(b_0 + \sum_{j=1}^{m} b_j \operatorname{sign}\left(a_{j0} + \sum_{i=1}^{d} a_{ji} x_i\right)\right),$$

where $m \leq p(d)$, all a_{ji}, b_j are integers of absolute value at most $p(d)$, and every displayed sign has a nonzero argument on every Boolean input.

This is the depth-two majority simulation of Goldmann and Karpinski [2]. Majority gates allow repeated and negated inputs and constants; collecting these wires gives the integer coefficients in the lemma. Their size convention counts wires, so their polynomial size bound also bounds all these coefficients. Their paper also gives the statement directly in terms of polynomially bounded integer weights. This is an external circuit theorem, not a new proof of threshold simulation here. A threshold function initially specified with a fixed convention for ties can be put in strictly separating form by an arbitrarily small bias change on the finite Boolean cube.

Lemma 4.2 (A polynomial shallow tanh realization)**.** *Every Boolean threshold function on d inputs has a one-hidden-layer tanh network with linear readout, global signed margin at least $3/2$, and both width and maximum absolute physical weight bounded by a universal polynomial in d. These bounds are independent of the coefficients and geometric margin of its original threshold representation.*

Proof. Use Lemma 4.1. Put $A = \max\{1, \sum_{j=1}^{m} |b_j|\}$ and $\lambda = \frac{1}{2}\log(8A)$. Each inner affine form is a nonzero integer, so, with $h_j(x)$ denoting its sign,

$$\left|\tanh\left(\lambda\left(a_{j0} + \sum_i a_{ji} x_i\right)\right) - h_j(x)\right| \leq 2e^{-2\lambda} = \frac{1}{4A}.$$

Consequently replacing all bottom signs by these tanh features changes the top affine form by at most $1/4$. Its exact signed value is a positive integer and hence at least one. Therefore

$$z(x) = 2b_0 + 2\sum_{j=1}^{m} b_j \tanh\left(\lambda\left(a_{j0} + \sum_i a_{ji} x_i\right)\right)$$

satisfies $f(x)z(x) \geq 3/2$ everywhere.

Implement the constant $2b_0$ by one additional hidden feature equal to $1/2$, using first-layer bias $\operatorname{atanh}(1/2)$ and readout coefficient $4b_0$. The resulting width is $m+1$, and every physical weight has magnitude at most

$$\max\{\lambda p(d), 4p(d), \operatorname{atanh}(1/2)\},$$

which is polynomial in d. No later bias is used. □

Theorem 4.3 (Unrestricted learning of every Boolean halfspace)**.** *Fix $\zeta > 0$. There is a polynomial P_ζ such that, for every Boolean threshold target f on d inputs, every learner depth $L \geq 2$, and every input distribution, a sample of size*

$$n \geq P_\zeta(d, L, \varepsilon^{-1}, \log(2/\delta))$$

has, with probability at least $1-\delta$, the following simultaneous guarantee. At every width $N \geq C_L(f)$ and every prior scale $0 < \sigma^2 \leq N^{-(1+\zeta)}$, the original unrestricted Gaussian posterior has Gibbs error at most ε. Here $C_L(f)$ is the actual bounded-weight network complexity at the specified learner depth, and the posterior is conditioned only on the original global clean output margin and the observed training labels.

In particular, the guarantee holds for every $\gamma \geq 1+\zeta$, with no dependence on an inverse geometric margin or on the precision of a specified affine representation of f.

Proof. Lemma 4.2 supplies polynomial-in-d values of the shallow width C and physical weight bound R in Theorem 22.1. Substitute these values in that theorem. Its smaller-width argument already covers every $C_L(f) \leq N < M$ below the polynomial-width constructed witness; no identification of $C_L(f)$ with that constructed width is required. □

This is an information-theoretic posterior guarantee. It does not assert that sampling the conditioned posterior, or recovering the alternative shallow representation from examples, is computationally efficient.

5 Targets with small Fourier spectral norm

For a Boolean function $f : \{-1,1\}^d \to \{-1,1\}$, write

$$f(x) = \sum_{S \subseteq [d]} \widehat{f}(S)\chi_S(x), \qquad \chi_S(x) = \prod_{i \in S} x_i, \qquad A(f) = \sum_S |\widehat{f}(S)|.$$

The Fourier coefficients here use the uniform distribution, even when the learning distribution is different. Parseval implies $A(f) \geq 1$. The following sufficient condition imposes neither a Fourier-degree bound nor a bound on the number of nonzero Fourier coefficients.

Lemma 5.1 (A shallow realization from Fourier spectral norm)**.** *Every such f, with $A = A(f)$, has a one-hidden-layer tanh realization with linear readout, signed margin at least $13/8$, width at most*

$$1 + d\left\lceil 128A^2(d+2)\log 2 \right\rceil,$$

and physical weights bounded by

$$\max\{3A, \tfrac{d}{2}\log(32Ad), 1\}.$$

Proof. Sample a set S with probability $|\widehat{f}(S)|/A$ and attach the sign $s_S = \operatorname{sign} \widehat{f}(S)$. The random function $As_S\chi_S$ has mean f and takes values in $[-A, A]$. For $m = \lceil 128A^2(d+2)\log 2 \rceil$, Hoeffding's inequality and a union bound over the 2^d inputs show that, with positive probability,

$$\left| \frac{A}{m} \sum_{j=1}^{m} s_{S_j} \chi_{S_j}(x) - f(x) \right| \le \frac{1}{8} \qquad \text{for every } x.$$

Fix one such choice.

Each parity on $k \ge 1$ variables admits an elementary real-output threshold expansion. Put $t = \sum_{i \in S} x_i$ and $\theta_j = -k + 2j - 1$ for $1 \le j \le k$. On the permitted values $t \in \{-k, -k+2, \dots, k\}$,

$$\chi_S(x) = b_k + \sum_{j=1}^{k} (-1)^{k-j} \operatorname{sign}(t - \theta_j), \qquad b_k \in \{0, 1\}.$$

Indeed, both sides start at $(-1)^k$ and have the same jump at each threshold; explicitly $b_k = 1$ for even k and zero for odd k. Every threshold has absolute affine margin at least one. With $\lambda = \frac{1}{2}\log(32Ad)$, replace each sign by $\tanh(\lambda(t - \theta_j))$. The resulting parity approximation has uniform error at most

$$2ke^{-2\lambda} \le \frac{1}{16A}.$$

The empty parity is the constant one. A single constant hidden neuron, whose output is $\tanh 1$, supplies all constant terms.

Replacing the m sampled parities by these approximations adds error at most $1/16$. Thus the resulting scalar output is within $3/16$ of f everywhere. Doubling the readout gives signed margin at least $13/8$. There are at most md nonconstant hidden neurons. Their first-layer coefficients and biases have magnitude at most $d\lambda$; their output coefficients have magnitude at most $2A/m$. The single constant neuron's output coefficient has magnitude at most $2A/\tanh 1 < 3A$. These are the stated bounds. □

Corollary 5.2 (Unrestricted learning from small spectral norm)**.** *For each fixed $\zeta > 0$, there is a polynomial P_ζ such that for every Boolean target f, every input distribution, and every learner depth $L \ge 2$, a sample size*

$$n \ge P_\zeta(d, A(f), L, \varepsilon^{-1}, \log(2/\delta))$$

gives, with probability at least $1 - \delta$, Gibbs error at most ε for the original unrestricted Gaussian posterior, simultaneously for every $N \ge C_L(f)$ and every $0 < \sigma^2 \le N^{-(1+\zeta)}$. The posterior retains its original conditioning on a global clean output margin and the training labels.

Proof. Substitute Lemma 5.1 into Theorem 22.1. All witness parameters are polynomial in d and $A(f)$. The all-width conclusion is the all-width conclusion of that theorem and does not require $C_L(f)$ to equal the width of the constructed shallow realization. □

Both the realization lemma and its learning corollary also hold with $A(f)$ replaced by $\sum_S |\widehat{g}(S)|$ for any real-valued Fourier polynomial g satisfying $f(x)g(x) \ge 1$ on every input. The same sampling argument approximates g to error $3/16$, which still leaves the stated signed margin after doubling the readout. Thus a small normalized coefficient sum for a margin-separated vote of parities is enough, even when the Boolean sign function itself has a larger Fourier norm.

For example, every parity has $A(f) = 1$, independently of its degree. More generally, a function with s nonzero Fourier coefficients has $A(f) \le \sqrt{s}$ by Cauchy–Schwarz and Parseval. These are sufficient representation conditions. The argument does not assert that all small Boolean formulas or all narrow branching programs have polynomial Fourier spectral norm.

Corollary 5.3 (Two-state branching programs)**.** *Every Boolean function computed by a deterministic layered width-two branching program of length T has $A(f) \le T+1$. Consequently the sample bound in Corollary 5.2 is polynomial in $d, T, L, \varepsilon^{-1}, \log(2/\delta)$ for this class. The program may query variables repeatedly, and its two nodes in a layer may query different variables.*

Proof. Encode the two states as $-1, +1$, and let $s_j(x)$ be the state after j transitions. Given the old state $+1$ or -1, the next state is a Boolean function $u_+(x)$ or $u_-(x)$ of at most one input bit. Each such function is constant or a signed input bit, so $A(u_+) = A(u_-) = 1$. The transition can be written

$$s_{j+1}(x) = a_j(x)s_j(x) + b_j(x), \qquad a_j = (u_+ - u_-)/2, \quad b_j = (u_+ + u_-)/2.$$

The Fourier coefficient norm is subadditive and submultiplicative: the latter follows because multiplication of parities uses symmetric difference of their index sets. Thus $A(a_j), A(b_j) \le 1$ and

$$A(s_{j+1}) \le A(s_j) + 1.$$

The initial state is constant, so $A(s_0) = 1$. The final output is either constant or a signed copy of the final state, proving the claim. Unused state slots can be filled arbitrarily. If the program uses absorbing states for early termination, these states count toward its width-two bound. □

The same proof allows a node to query a parity rather than a single bit, since a signed parity also has Fourier norm one. Decision lists with T parity tests likewise have Fourier norm at most $T + 1$, by a separate direct argument: write the output as the sum of its $T + 1$ signed first-accept and default indicators. Each indicator is a product of factors $(1 \pm \chi_S)/2$, each of Fourier norm at most one, so submultiplicativity and the triangle inequality give the bound. These extensions concern representation and sample bounds only; no efficient algorithm for identifying a program from samples is asserted.

6 Constant-depth threshold circuits and a lower-bound barrier

The coefficient-independent halfspace construction also clarifies which explicit targets could disprove the general unrestricted sample conjecture. Here a threshold circuit may have arbitrary real weights, and its size counts wires, including input and constant wires.

Lemma 6.1 (Threshold circuits as bounded-weight tanh networks)**.** *There is a universal constant a such that every depth-h, $h \ge 1$, threshold circuit of wire size at most s on d Boolean inputs has a tanh realization with linear readout, unit global margin, depth at most $2h$, and both width and maximum absolute physical weight bounded by $(2 + d + s)^a$. It uses no biases after the augmented input. The bound is independent of the original real coefficients and of every original gate's geometric margin.*

Proof. Replace each threshold gate by its depth-two polynomial-size, polynomial-integer-weight simulation from Lemma 4.1. Each gate has at most s inputs. The total wire count, coefficient bound, and maximum weighted fan-in of the resulting circuit are bounded by a universal polynomial in $d + s$. Its depth is at most $2h$. Subdivide skip wires by identity threshold gates to make the circuit layered; since $h \le s$, this costs only another polynomial factor. Pad with identity gates as needed so that the output is at a common depth $q = 2h \ge 2$.

Every gate can be taken to have integer affine form with absolute value at least one on Boolean inputs. Let $B \ge 1$ bound the sum of the absolute coefficients of any such affine form, including its constant. Put $\delta = 1/(4B)$ and $\lambda = 2\log(8B)$. Replace every gate except the final output gate by tanh applied to λ times its affine form. If all preceding features approximate their intended signs to

accuracy δ, the affine perturbation is at most $B\delta = 1/4$. Its signed value is therefore at least $3/4$, and the new feature error is at most

$$2e^{-3\lambda/2} = \frac{2}{(8B)^3} \le \delta.$$

This inductive accuracy bound does not deteriorate with depth.

To implement constants without later biases, add a constant feature equal to $1/2$ at every hidden layer. The first such feature has bias $\operatorname{atanh}(1/2)$; each later one receives weight $2\operatorname{atanh}(1/2)$ from the preceding constant feature. An affine constant b is then supplied by weight $2\lambda b$ to this feature. Thus all constants are exact. The last threshold is replaced by twice its affine form, as a linear readout; its signed value is at least $3/2$. Uniform zero-padding of the hidden layers preserves the function. Every used parameter is polynomial in $d + s$, as asserted. □

Corollary 6.2 (Precision-independent threshold-circuit learning)**.** *Fix $\zeta > 0$. A target with a depth-h, wire-size-s threshold circuit has unrestricted Gaussian posterior sample complexity*

$$(2 + d + s)^{O_\zeta(h)}\big(L + \varepsilon^{-1} + \log(2/\delta)\big)^{O_\zeta(1)}$$

at every learner depth $L \ge 2h$, simultaneously for every $N \ge C_L(f)$ and every $0 < \sigma^2 \le N^{-(1+\zeta)}$. No original coefficient magnitude or geometric margin enters the bound. In particular, for each fixed h, this is polynomial in the circuit size, input dimension, learner depth, and statistical parameters.

Proof. Apply the fixed-intrinsic-depth theorem and its quantitative corollary below to Lemma 6.1, with intrinsic depth $D_0 = 2h$ and polynomial bounds on the tanh width and weights. □

Recall that nonuniform TC^0 consists of Boolean families having polynomial-size threshold circuits of some fixed depth. Consequently every such family enjoys a polynomial sample guarantee under the present posterior at all sufficiently large learner depths. This gives the following precise limitation on a negative-result route.

Corollary 6.3 (A circuit-complexity consequence of a counterexample)**.** *Fix $\zeta > 0$. Suppose a family (f_d), at learner depths $L(d) \to \infty$, requires more than every polynomial in $d, L(d)$ examples for some fixed positive error and confidence targets, under the present unrestricted posterior and the all-admissible-width guarantee with $\gamma \ge 1 + \zeta$. Then $(f_d) \notin \mathsf{TC}^0$. In particular, a family in NC^1 with this property would separate NC^1 from TC^0.*

Proof. If the family were in TC^0, it would have fixed depth h and wire size polynomial in d. Eventually $L(d) \ge 2h$, so Corollary 6.2 would supply a polynomial sample bound simultaneously for every admissible width and stated Gaussian scale, a contradiction. □

This is a logical consequence, not a claimed circuit separation or a proof that unrestricted learning is always polynomial. It also does not cover exponents approaching one with dimension, fixed learner depths below the alternative representation's depth, or the mean-field case.

7 Finite hidden-state bottlenecks give polynomial depth bounds

Boolean inputs permit a positive result that has no analogue for a continuously varying scalar input. If a short initial part of the teacher produces only polynomially many distinct hidden states, its remaining computation can be replaced by a lookup circuit. The remaining teacher layers may be arbitrarily expansive. No additional condition is imposed on the learner's prior.

Proposition 7.1 (A finite first-layer state set)**.** *Suppose a tanh network in the present architecture computes $f : \{-1,1\}^d \to \{-1,1\}$ and its first hidden vector has C coordinates and at most K distinct values on the cube. Then f has a depth-three threshold circuit with $O(KC(d+1))$ wires. Consequently it has a globally unit-margin tanh realization of depth six, with width and individual weights polynomial in d, C, K. There is no bound on the depth, coefficients, or numerical stability of the specified teacher's remaining computation.*

Proof. For every attained hidden vector v, its label is well-defined because all subsequent computation depends only on that vector. Tanh is injective. Thus the condition that its jth coordinate equals v_j can be tested by requiring the original affine preactivation to lie between two thresholds that isolate its corresponding value from all other attained preactivation values. There are only finitely many such values. Choose separating endpoints strictly between consecutive values; at an extreme, omit the unnecessary test or use a constant predicate.

Equality to v is therefore a conjunction of at most $2C$ affine threshold predicates of the original input. Use one threshold layer for these predicates, a second for each state conjunction, and a final OR over the positively labeled states. There are at most K states and each first-layer predicate uses at most $d+1$ wires. Constant functions are immediate. Lemma 6.1 gives the stated bounded-weight tanh realization. In particular, arbitrarily small gaps between attained real values cause no coefficient-size loss in the resulting realization. □

The proposition already gives a sample polynomial in d, C, K, L and the statistical parameters at every learner depth $L \geq 6$, simultaneously for all $N \geq C_L(f)$ and all $0 < \sigma^2 \leq N^{-(1+\zeta)}$, for each fixed $\zeta > 0$. The next result allows a bottleneck after any fixed number of teacher layers. Its proof uses binary precision, rather than a lower bound on the gaps between all attained states.

Lemma 7.2 (Fixed-depth tanh evaluation by threshold circuits)**.** *Fix $j \geq 1$. Consider the first j hidden layers of a tanh network with widths at most C, input $(1,x)$ for $x \in \{-1,1\}^d$, and weights of magnitude at most $B \geq 1$. For every integer $p \geq 1$, a threshold circuit of depth depending only on j and wire size polynomial in d, C, B, p produces a dyadic vector $\widehat{h}(x)$ such that*

$$\sup_x \|\widehat{h}(x) - h^j(x)\|_\infty \leq 2^{-p}.$$

The polynomial degree may depend on j. Its advice consists of polynomially many bits of the specified real parameters; no computability assumption on those real parameters is required.

Proof. We use the theorem of Hesse–Allender–Barrington [3]: division and iterated multiplication of polynomially many polynomial-bit integers have polynomial-size constant-depth threshold circuits. Addition, iterated addition, and comparison also have such circuits. Only their nonuniform consequence is needed here.

First consider a dyadic scalar $t = a/2^b$ with $|t| \leq R$, $R \geq 1$. Tanh can be approximated to error 2^{-p} by a constant-depth threshold circuit of size polynomial in b, R, p. Indeed, take the degree-m Taylor polynomial of e^{2t}, with $m = \lceil 40(R+p+2) \rceil$. Its uniform remainder is at most

$$e^{2R}\frac{(2R)^{m+1}}{(m+1)!} \leq e^{2R}\left(\frac{2eR}{m+1}\right)^{m+1} < 2^{-p-4}.$$

The polynomial is the exact rational number U/V, where

$$V = 2^{bm}m!, \qquad U = \sum_{k=0}^{m}(2a)^k 2^{b(m-k)}\frac{m!}{k!}.$$

All products and sums involve polynomially many integers with $O(m(b+\log(R+1)+\log(m+1)))$ bits. Negative summands are handled exactly; cancellation does not introduce numerical error. The quoted arithmetic theorem computes these quantities at constant depth. Since U/V is within 2^{-p-4} of a nonnegative number, $U/V+1 \geq 3/4$. Hence

$$\frac{U-V}{U+V}$$

approximates $(e^{2t}-1)/(e^{2t}+1) = \tanh t$ to error less than 2^{-p-2}. A final rational division and dyadic rounding give the required error. Clamping the result to $[-1,1]$ can only decrease its distance from $\tanh t$.

To evaluate the prefix, round each parameter to absolute error 2^{-q} and approximate every tanh to the same error, where

$$q = p + \left\lceil \log_2\bigl(4j(d+C+2)(2CB)^j\bigr) \right\rceil.$$

Clamp all approximate hidden coordinates to $[-1,1]$. If E_s is the maximum hidden-coordinate error after layer s, then

$$E_1 \leq (d+2)2^{-q}, \qquad E_s \leq CBE_{s-1} + (C+1)2^{-q} \quad (2 \leq s \leq j).$$

Therefore

$$E_j \leq j(d+C+2)(2CB)^j 2^{-q} \leq 2^{-p}.$$

The rounded preactivations have magnitude at most $(d+C+1)(B+1)$ and polynomial bit length. Every affine calculation uses only a constant number of the arithmetic operations above, and each tanh uses the preceding scalar construction. Composing the fixed number j of stages preserves constant threshold depth and polynomial wire size. The rounded matrices have polynomial total description length. Their bits are fixed circuit advice, rather than quantities the circuit is required to extract from arbitrary real numbers. □

Theorem 7.3 (A finite bottleneck after a fixed prefix)**.** *Fix $j \geq 1$ and $\zeta > 0$. Suppose a width-C, depth-$D > j$ tanh teacher has weights bounded by $B \geq 1$, signed output at least $1/2$ on the Boolean cube, and at most K distinct vectors after its jth hidden layer. There are a constant q_j and a polynomial p_j, with degrees depending only on j, such that f has a globally unit-margin tanh realization of depth q_j, width and weights at most $p_j(d,C,B,K,D)$.*

Consequently the original unrestricted Gaussian posterior has sample complexity polynomial in

$$d, C, B, K, D, L, \varepsilon^{-1}, \log(2/\delta)$$

at every learner depth $L \geq q_j$, simultaneously for every $N \geq C_L(f)$ and every $0 < \sigma^2 \leq N^{-(1+\zeta)}$. The same polynomial form holds at every learner depth $L \geq D$, including the possible bounded-depth range below q_j. The constants and polynomial degrees may depend on j and ζ, but not on the remaining teacher depth $D-j$.

Proof. In the infinity norm, the teacher suffix from its jth hidden layer to the linear readout is Lipschitz with constant at most

$$S = (CB)^{D-j}.$$

Oppositely labeled inputs have outputs differing by at least one. Consequently their jth hidden states are separated in infinity norm by at least S^{-1}. Set

$$p = \lceil \log_2(32S) \rceil, \qquad \eta = 2^{-p}.$$

The required precision is only $O(1 + D\log(2CB))$ bits, despite the possibly exponentially small separation.

Apply Lemma 7.2 to produce $\widehat{h}(x)$ within η of the true prefix state. For each attained state v, hardcode a dyadic center $\widehat{v}$ within η of v, and hardcode its Boolean label. Output $+1$ exactly when $\widehat{h}(x)$ lies within 3η in every coordinate of at least one positive center. A positive input lies within 2η of its own center. A negative input is at distance at least $S^{-1} - 2\eta \geq 30\eta$ from every positive center. Thus the lookup returns the correct label everywhere.

There are at most K centers, each described by $O(Cp)$ bits. Subtraction, comparison, a conjunction over coordinates, and a disjunction over centers are constant-depth threshold operations of polynomial size. Together with the fixed-depth prefix evaluation, this gives a threshold circuit of depth depending only on j and size polynomial in d, C, B, K, D. Apply Lemma 6.1 and pad the resulting fixed depth to a constant q_j. This proves the representation statement.

The fixed-intrinsic-depth learning theorem gives the claimed all-admissible-width guarantee for $L \geq q_j$. To cover every $L \geq D$, it remains only to consider $D < q_j$ and $L < q_j$. There are finitely many such intrinsic teacher depths, bounded in terms of j alone. Double the supplied teacher's readout and apply the same fixed-intrinsic-depth theorem directly to that teacher. Taking the maximum of these finitely many polynomial bounds gives the stated form. The prior throughout retains only its original global output margin restriction and the observed training signs. □

In the representation proof it would suffice to cover the attained prefix states by K infinity-norm balls of radius η, with centers chosen among those states. Use boxes of radius 4η in the lookup. Opposite labels cannot occur in the same such ball, and the same separation argument applies. Exact finite-state count is a particularly simple sufficient condition; no separation of same-label states is needed.

Corollary 7.4 (Discrete input statistics)**.** *The following teachers satisfy Proposition 7.1.*

1. *If the nonconstant entries of first-layer row j are A_{ji}/Q_j, with integers A_{ji} and $Q_j > 0$, put $R_j = \sum_{i=1}^{d} |A_{ji}|$. Its state count is at most*

$$K \leq \prod_{j=1}^{C} (2R_j + 1).$$

   Its first-layer biases may be arbitrary real numbers. In particular, first-layer input weights bounded by B with common denominator Q give $K \leq (2dBQ + 1)^C$. For fixed C and polynomially bounded B, Q, the sample bound at every learner depth $L \geq 6$ is polynomial in input dimension and learner depth, independently of all later teacher computation.

2. *If the first-layer matrix has at most k distinct nonconstant input-column vectors, let $d_1, \ldots, d_k$ be their multiplicities. Then*

$$K \leq \prod_{a=1}^{k} (d_a + 1) \leq (d + 1)^k.$$

   For fixed k, at every learner depth $L \geq 6$ the sample bound is polynomial in d, C, L and the statistical parameters, with no dependence on the precision or magnitude of the specified teacher coefficients. Functions of a fixed number of coordinate-block Hamming weights are examples.

Proof. In the first case, the integer sum in row j lies between $-R_j$ and R_j, so it has at most $2R_j + 1$ values. A fixed bias and tanh do not increase this count. In the second case, the first affine map

depends only on the sum of the Boolean inputs in each column group. A group of size d_a has exactly $d_a + 1$ possible sums. All later computation factors through the resulting state vector. Apply the proposition and the fixed-intrinsic-depth learning theorem. □

These results use finite information passed by a teacher prefix, rather than stability of the prior's networks. They do not imply a joint polynomial theorem for an arbitrary growing prefix depth or unrestricted state count. Even one bounded real affine projection can take 2^d distinct values on the cube. Rounding that projection while preserving a long suffix's output signs may require exponentially fine precision in depth, so it does not establish the required polynomial state bound for a general teacher.

8 A depth-blocking bound for general Boolean circuits

A simple circuit transformation slightly improves the exponential dependence on growing intrinsic depth. It does not make the dependence polynomial for general small circuits.

Proposition 8.1 (Blocking Boolean layers)**.** *Let f be computed by an AND/OR/NOT circuit of fan-in at most two, depth $h \geq 1$, and at most s gates and input wires. Put*

$$S = \max\{16, 2 + d + s\}, \qquad b = \max\{1, \lfloor \log_2 \log_2 S \rfloor\},$$
$$a = \lfloor h/b \rfloor, \qquad r = h - ab, \qquad q = \max\{2, 2a + \min\{2, r\}\}.$$

Then f has a depth-q tanh representation with unit global margin, width and physical weights bounded by a universal polynomial in S. Consequently, for each fixed $\zeta > 0$, at every learner depth $L \geq q$ the unrestricted posterior has a sample bound of the form

$$S^{O_\zeta(1+h/\log\log S)}\big(L + \varepsilon^{-1} + \log(2/\delta)\big)^{O_\zeta(1)},$$

uniformly for every actual width $N \geq C_L(f)$ and every $0 < \sigma^2 \leq N^{-(1+\zeta)}$. One has $q \leq \max\{2, h\}$, so in particular the result applies at every learner depth $L \geq \max\{2, h\}$. For polynomial-size logarithmic-depth circuits, this is $\exp[O_\zeta(\log^2 S/\log\log S)]$ times a polynomial in the learner depth and statistical parameters.

Proof. First insert identity gates along skip wires and at an output of depth less than h. The resulting layered circuit has polynomial size in S, since $h \leq s$ after removing unused gates. Divide its layers into consecutive blocks of at most b layers.

Each gate at the end of a block depends on at most 2^b signals at the beginning of that block. Regard its subcomputation as a Boolean function of these boundary signals, and express that function by its full DNF. The DNF has at most

$$2^{2^b} \leq S$$

terms, each of length at most $2^b \leq \log_2 S$. Implementing its conjunctions in one threshold layer and its final disjunction in the next uses polynomially many wires and integer coefficients bounded by a polynomial in S. A gate whose function is constant can be implemented by a constant threshold. Use two threshold layers for every full block. A final one-layer block can simply retain its original AND/OR/NOT threshold gates, rather than be expanded to two layers; a longer final partial block uses two layers. The resulting threshold depth is $2a + \min\{2, r\} \leq h$. When $h = 1$, pad to depth two by inserting one identity threshold layer. All blocks together therefore form a depth-q threshold

circuit of polynomial size and polynomial integer weights. Choose odd integer thresholds after doubling literal coefficients, so every gate has signed preactivation at least one on Boolean inputs.

Here the threshold circuit already has small integer weights; no threshold-to-majority replacement is needed. Convert its first $q-1$ layers directly to tanh. If $B \geq 1$ bounds each gate's total absolute incoming coefficients, including its bias, use gain $\lambda = 2\log(8B)$ and accuracy $1/(4B)$ at every hidden layer. Inductively the affine perturbation from preceding approximate signs is at most $1/4$, and the tanh error is at most $2e^{-3\lambda/2} \leq 1/(4B)$. Biases are represented by an exact hidden constant $1/2$, propagated using weight $2\operatorname{arctanh}(1/2)$; its contribution is included in B. At the last threshold, use twice the affine form as the linear readout. It has signed value at least $3/2$. This construction has exactly q weighted layers, no later biases, and polynomial width and weights.

The quantified fixed-intrinsic-depth theorem then gives a sample bound $S^{O_\zeta(q)}$ times the displayed polynomial in L and the statistical parameters. Since $q = O(1 + h/\log\log S)$, this is the claimed estimate. Its small-width part is already expressed using the actual complexity $C_L(f)$, so the alternative circuit's larger width does not impose an additional lower-width assumption. □

This is a circuit-size/depth tradeoff rather than a polynomial-product representation theorem. In particular, for balanced recursive majority it improves the available quasipolynomial upper bound but does not settle whether a polynomial bound is possible.

9 Formula targets independently of their original depth

Formula balancing is classical, following Spira [4]. We include the elementary construction needed here to make the resulting sample bound explicit. This section does not claim a new circuit-balancing theorem.

Lemma 9.1 (Polynomial-size balancing of a Boolean formula)**.** *An AND/OR/NOT formula with s gates and literal occurrences has an equivalent fan-in-two circuit of size polynomial in $s+2$ and depth $O(\log(s+2))$, regardless of the original formula depth.*

Proof. Push negations to the leaves using De Morgan's laws, eliminate double negations, and simplify constants. This does not increase the number of literal occurrences. Binarize any larger-fan-in AND/OR gates; this adds only linearly many gates and does not change the leaf count. A constant formula is immediate. Otherwise let n denote its number of literal occurrences, counting repeated occurrences separately.

For $n \geq 2$ there is a subformula g with between $n/3$ and $2n/3$ literal occurrences. To find one, descend from the root while a child has more than $2n/3$ leaves. At the last such node both children have at most $2n/3$ leaves, and at least one has at least $n/3$ leaves. Regard this one occurrence of g as a formal Boolean input. After substituting either value $b \in \{0,1\}$ and simplifying constants, the remaining formula F_b has at most $n - |g| \leq 2n/3$ literal occurrences. The Shannon identity is

$$F = (g \wedge F_1) \vee (\neg g \wedge F_0).$$

It is valid even if variables appearing in g also occur elsewhere in the original formula.

Recursively balance g, F_0, F_1 and join the three resulting circuits by the four indicated Boolean gates. Their depths increase by at most three, and each recursive subproblem has at most $2n/3$ occurrences. Hence the depth is $O(\log(n+1))$. There are at most three recursive subproblems per call, so the size is polynomial; for example a bound $O((n+1)^4)$ follows immediately by induction. Reuse the balanced output of g in its two appearances in the displayed identity. This establishes the asserted circuit bounds. □

Corollary 9.2 (A quasipolynomial bound for arbitrary formulas)**.** *Let f have a Boolean formula with at most s gates and literal occurrences, and put $S = \max\{16, 2 + d + s\}$. There is a tanh realization with unit margin, polynomial-in-S width and physical weight bounds, and depth q satisfying*

$$2 \le q \le C \frac{\log S}{\log \log S}$$

for a universal constant C. For each fixed $\zeta > 0$ and every learner depth $L \ge q$, the unrestricted posterior sample bound is

$$\exp\left[O_\zeta\left(\frac{\log^2 S}{\log \log S}\right)\right] \left(L + \varepsilon^{-1} + \log(2/\delta)\right)^{O_\zeta(1)},$$

simultaneously for all actual widths $N \ge C_L(f)$ and all $0 < \sigma^2 \le N^{-(1+\zeta)}$. This applies in particular to read-once formulas and does not depend on the depth of the given formula.

Proof. Lemma 9.1 gives a circuit of polynomial size and depth $O(\log S)$. Apply Proposition 8.1 to that circuit. The logarithm of its size parameter is $O(\log S)$ and its double logarithm is comparable to $\log \log S$. Its blocked tanh depth is therefore $O(\log S / \log \log S)$, with polynomial width and weights. The quantified intrinsic-depth bound gives the displayed sample estimate. All-width coverage is inherited from that bound, not from identifying the target's actual minimum width with the construction. □

When a supplied formula is already shallower, its original blocked representation can be used instead. The result above is still quasipolynomial in formula size; it does not resolve polynomial sample complexity for all polynomial-size formulas.

10 Polynomial product norm suffices after logarithmic added depth

The posterior representation ceiling has a partial converse for a specified finite-width target representation. Polynomial product norm alone does not balance the physical tanh layers, but small-prefix linearization and unequal clone counts balance all retained hidden layers. Only a polynomial readout cost then needs to be transmitted.

Theorem 10.1 (Deep targets with polynomial product norm)**.** *Suppose a width-C, depth-$D_0 \ge 2$ tanh network has global signed margin at least one for f, and product of layer Frobenius norms at most P. Put $P_0 = \max\{1, P\}$ and*

$$\tau^2 = \frac{1}{16 P_0 D_0 (d+1)^{3/2}}, \qquad T = \frac{32 C P_0^2}{\tau^2}.$$

For every fixed $\zeta > 0$, the original unrestricted Gaussian posterior has sample complexity polynomial in

$$d, C, D_0, P_0, L, \varepsilon^{-1}, \log(2/\delta)$$

at every learner depth

$$L \ge D_0 + \lceil \log T \rceil,$$

simultaneously for every $N \ge C_L(f)$ and every $0 < \sigma^2 \le N^{-(1+\zeta)}$. The same polynomial conclusion holds under the alternative depth condition $L \ge 2D_0$. There is no bound on the individual weights of the specified target representation and no internal robustness restriction.

Proof. Write M_j for the actual Frobenius norm of its jth matrix, including the readout, and $S_j = \prod_{i \le j} M_i$ for the prefix products. All norms are positive, since the global margin excludes a zero map. The augmented input norm is $R_0 = \sqrt{d+1}$. At every hidden layer, boundedness of tanh and the suffix Lipschitz bound give

$$1 \le |z(x)| \le \sqrt{C}\,\frac{P}{S_j}, \qquad S_j \le P\sqrt{C}.$$

Using an upper bound P instead of the exact full product only weakens this inequality.

Replace the tanh activation by the identity at each hidden layer whose original prefix product satisfies $S_j < \tau$. The inequality $|u - \tanh u| \le |u|^3/3$ and a telescoping argument give total output change at most

$$\sum_{j:S_j<\tau} \frac{P}{S_j}\frac{(S_j R_0)^3}{3} \le \frac{P D_0 \tau^2 R_0^3}{3} \le \frac{1}{48}.$$

The same prefix and suffix bounds apply throughout this telescoping, since both identity and tanh are zero-preserving and one-Lipschitz. Collapse the resulting consecutive linear matrices into their next nonlinear layer or the final readout, then double the readout. The result still computes f with unit margin.

If every hidden activation was removed, f is a Boolean halfspace, and Theorem 4.3 proves the claim. Otherwise let $H \ge 1$ be the number of retained hidden layers, with original indices $j_1 < \cdots < j_H$. Their retained prefix bounds satisfy

$$\tau \le S_{j_s} \le P_0\sqrt{C}.$$

The first collapsed matrix has Frobenius norm at most S_{j_1}; the sth has norm at most $S_{j_s}/S_{j_{s-1}}$; and the doubled readout has norm at most $2P_0/S_{j_H}$. These are assigned norm bounds; matrix multiplication can make the actual norms smaller without affecting any subsequent upper bound. In particular every collapsed physical weight is at most

$$R = \max\{1, 2P_0\sqrt{C}/\tau\}.$$

Define

$$a = \frac{d\tau^2}{8C}, \qquad r_s = \frac{a}{dS_{j_s}^2} = \frac{\tau^2}{8CS_{j_s}^2}, \qquad r_{\min} = \frac{\tau^2}{8C^2P_0^2}.$$

Then $r_{\min} \le r_s \le 1/(8C)$ and $a < 1/8$. At common width N, give every original neuron in retained layer s exactly $c_s = \lceil N r_s \rceil$ clones, divide outgoing weights by c_s, and fill the remaining positions with zero neurons. For

$$N \ge M := \lceil \max\{2C, 8D_0/r_{\min}, 2R, 4T, 4\} \rceil,$$

the clone counts fit in width N and $Nr_s \le c_s \le Nr_s(1 + 1/(8D_0))$. This preserves the collapsed network exactly. Its pre-readout energy and readout energy obey

$$E_{\text{pre}} \le a\left(1 + \frac{1}{8D_0}\right) + (H-1)\left(1 + \frac{1}{8D_0}\right) \le H,$$

$$c^2 = N\|v_N\|_2^2 \le \frac{dS_{j_H}^2}{a}\frac{4P_0^2}{S_{j_H}^2} = \frac{32CP_0^2}{\tau^2} = T.$$

For the first inequality, $H \le D_0 - 1$ and $a < 1/8$ leave more than enough slack to absorb all clone rounding.

Replace the readout by the scalar-transmission construction with $m = L - H$ weighted layers, and double its final readout as in the shallow-target proof. The resulting unit-margin witness has energy

$$K \le H + (m + 15)T^{1/m}.$$

The stated depth conditions ensure $m \ge 2$. If $L \ge D_0 + \lceil \log T \rceil$, then $m \ge \log T$, and $e^u \le 1 + 2u$ for $0 \le u \le 1$ gives

$$K \le L + 15 + 32 \log T.$$

Hence the Gaussian-distance parameter is bounded by

$$Q_*^2 = ((K + 2)/L)^L \le e^{17} T^{32}.$$

Under the alternative condition $L \ge 2D_0$, one has $L/m \le 2$ and the direct estimate

$$K + 2 \le (L + 17)T^{1/m}, \qquad Q_*^2 \le e^{17} T^2.$$

In either case, K, Q_* and M are polynomial in the stated target parameters and learner depth. The Gaussian-distance theorem therefore gives the required sample bound for all $N \ge M$.

We check the actual smaller-width convention. The collapsed weights are at most $R \le M/2$, and cloning can only reduce internal and readout weights. The first scalar-transmission matrix has weights at most $1.1R < M$. If $q^2 = \max\{c^2, 1\}^{1/m} \le T$, the remaining scalar weights have magnitudes at most $2.2q/N$ and $2q/N$. Since $M \ge 4T \ge 4$, these are less than M. Thus the constructed width-M target is admissible and $C_L(f) \le M$. For $C_L(f) \le N < M$, clone a defining minimum-width target and double its readout; its width and weights are bounded by a polynomial in M. The finite-width theorem supplies the same polynomial sample form. Intersect the two sample events, with failure probabilities $\delta/2$ each, to conclude. □

The theorem is polynomial in P itself. For a generic bounded-weight deep circuit, P may still grow exponentially with its depth. Thus this does not settle the unrestricted general small-circuit conjecture. It does show that a polynomial product-norm representation is a sufficient target condition after a logarithmic number of additional layers, without imposing that condition on the support of the prior.

11 Width compression with controlled product norm

The width dependence in Theorem 10.1 can be removed. Boundedness and near-linearity of tanh permit a norm-preserving compression: very large normalized rows can be dropped, very small rows can be linearized, and the remaining neurons can be sampled with a probability chosen to balance their two adjacent norms.

Theorem 11.1 (Finite-set width compression for tanh)**.** *Let X be a nonempty finite set of inputs of norm at most $R \ge 1$. Suppose a depth-$D_0 \ge 2$ tanh network with linear readout, arbitrary finite hidden widths, and no later biases has product of Frobenius norms at most $P_0 \ge 1$. For every $0 < \epsilon < 1$, it has a same-depth approximation $\widetilde{z}$ such that*

$$\max_{x \in X} |z(x) - \widetilde{z}(x)| \le \epsilon, \qquad \prod_{j=1}^{D_0} \|\widetilde{W}^j\|_F \le 2P_0,$$

and every new hidden width is bounded by a universal polynomial in

$$P_0, D_0, R, \log(2|X|), \epsilon^{-1}.$$

The bound is independent of the original widths and individual weights. An augmented constant input is allowed and counts toward R.

Proof. Put $H = D_0 - 1$, $P_* = 2P_0$, $e = \epsilon/(8H)$ and $\eta = 1/(4H)$. We compress the hidden layers in order from the first to the last. At each step the product norm increases by at most $1 + \eta$, and the output changes by at most $6e$ on X. Thus every intermediate product is below P_*, and the final product is at most $e^{1/4}P_0 < 2P_0$, while total error is less than ϵ. A zero matrix makes the entire network zero and is trivial, so all prefix and pair norms used below may be assumed positive. If a compression step produces a zero matrix, stop and use the zero network with width one; the accumulated approximation estimates still apply.

Consider a hidden matrix W and its immediately following matrix V; the latter may be the final linear readout. Let S be the product of all Frobenius norms strictly before W, with $S = 1$ at the first layer, and put

$$A = \|W\|_F \|V\|_F.$$

The incoming feature vector $h(x)$ has norm at most SR. The part strictly after V has Lipschitz constant at most $P_*/(SA)$. If w_i^T is row i of W and v_i is column i of V, write $a_i = \|w_i\|_2$ and $b_i = \|v_i\|_2$. Zero rows or zero outgoing columns can be deleted.

Set

$$B = P_*/e, \qquad \tau^2 = 3e/(P_* R^3), \qquad \Delta = e/(P_* R), \qquad Q = (B/\tau)^2.$$

Partition neurons into large, small and middle according as $Sa_i > B$, $Sa_i < \tau$, or $\tau \le Sa_i \le B$. There are at most $S^2\|W\|_F^2/B^2$ large neurons. Their activation vector has norm at most the square root of this count. Deleting them therefore changes the final output by at most $P_*/B = e$.

For the small neurons, replace tanh by the identity. Since $|u - \tanh u| \le |u|^3/3$, their combined activation error has norm at most $\tau^2 R^3 S\|W\|_F/3$. Propagation through the suffix bounds the output error by $P_*\tau^2 R^3/3 = e$. Their resulting contribution to the next preactivation is the linear map

$$B_0 = \sum_{i \text{ small}} v_i w_i^T, \qquad \|B_0\|_* \le \sum_{i \text{ small}} a_i b_i,$$

where $\|\cdot\|_*$ is the nuclear norm.

Let $A_m = \sum_{i \text{ middle}} a_i b_i$. If $A_m < \Delta A$, delete all middle neurons: their output contribution is at most $P_* R A_m/A \le e$. Otherwise $A_m \ge \Delta A > 0$, and we approximate their contribution by sampling. Define

$$Z = \sum_{i \text{ middle}} b_i/a_i, \qquad p_i = (b_i/a_i)/Z.$$

Draw m independent indices from (p_i). A sampled neuron keeps its incoming row w_i^T and has outgoing column $v_i/(mp_i)$. If $\widehat{A} = (Z/m)\sum_{r=1}^m a_{i_r}^2$ and $c = m/Z$, its new incoming and outgoing squared Frobenius norms are exactly

$$c\widehat{A} \quad \text{and} \quad \widehat{A}/c.$$

Moreover $\mathbb{E}\widehat{A} = A_m$. Since middle row norms lie between τ/S and B/S, one has

$$Z \max_i a_i^2 \le A_m Q \le AQ.$$

The random vector $v_i \tanh(w_i^T h(x))/p_i$ has norm at most $AQSR$, uniformly on X.

Consequently some choice of the samples has both

$$\widehat{A} \le A_m + \eta A$$

and middle-preactivation approximation error at most eSA/P_* simultaneously for every $x \in X$, provided

$$m \geq 64Q^2 \max\{\eta^{-2}, (P_*R/e)^2\} \log(8(|X|+1)).$$

Here and below take integer ceilings. For detail, the scalar random variables Za_i^2/A are bounded by Q, so Hoeffding controls their mean. For a Hilbert-valued random vector bounded by $AQSR$, the mean deviation has expectation at most $AQSR/\sqrt{m}$; bounded differences gives a dimension-free tail. A union bound over X and the scalar condition has positive success probability with the displayed constant. Propagation of the vector error gives final output error at most e.

We next compress the linear branch. Truncate the singular-value decomposition of B_0 to rank

$$k = \lceil P_*R/e \rceil,$$

obtaining $B_k = UV_0$ with $\|U\|_F^2 = \|V_0\|_F^2 = \nu := \|B_k\|_* \leq \|B_0\|_*$. The residual operator norm is at most $\|B_0\|_*/(k+1) \leq A/(k+1)$, so its final output error is at most $P_*R/(k+1) \leq e$. If $\nu = 0$, omit the linear branch entirely.

When the middle branch was sampled, retain $c = m/Z$. When it was deleted and $\nu > 0$, choose $c = 1/(\nu S^2)$ instead. Represent B_k approximately using K copies of each of its at most k singular channels: their incoming rows are the rows of $\sqrt{c/K}\, V_0$, repeated K times, and their outgoing columns are the corresponding columns of $U/\sqrt{cK}$. Their total incoming and outgoing squared norms are $c\nu$ and ν/c. Their linear limit is exactly $B_k h$. The tanh linearization error at the next preactivation has norm at most

$$\frac{c\nu^2(SR)^3}{3K}.$$

After propagation this is at most $P_* c\nu S^2 R^3/(3K)$, using $\nu \leq A$. In the sampled case,

$$Z \geq A_m S^2/B^2, \qquad c\nu S^2 \leq mB^2/\Delta;$$

in the deleted case this quantity equals one. Hence it suffices to use

$$K = \left\lceil \frac{P_* R^3}{e} \max\{1, mB^2/\Delta\} \right\rceil.$$

The linear-channel approximation then contributes at most e error. When there is no sampled middle branch, the same bound is harmless; one may use the displayed deterministic value of m merely to define a common width upper bound.

Combining the sampled and linear branches gives pair product norm

$$\widehat{A} + \nu \leq A_m + \sum_{i \text{ small}} a_i b_i + \eta A \leq (1+\eta)A.$$

The last inequality is Cauchy–Schwarz over the original rows and columns. If the middle branch was deleted, the pair product is simply $\nu \leq A$. These are exact norm calculations for the constructed tanh matrices, not only bounds on their linear limits.

The compressed hidden width is at most $m + kK$. All quantities in this bound are universal polynomials in P_0, H, R, e^{-1} and $\log(2|X|)$. Crucially, no original dimension of either adjacent hidden layer occurs. Only the two adjacent matrices are altered. Processing the next hidden layer changes its own width without undoing the width bound for the preceding one. Uniform zero-padding at the end makes all hidden widths equal if desired. Summing the at most six errors and multiplying the pair-product bounds proves the theorem. □

Corollary 11.2 (Uniform compression on a bounded input set)**.** *If the input dimension is q and X is any subset of the Euclidean ball of radius $R \geq 1$, the same conclusion holds with width polynomial in $q, P_0, D_0, R, \epsilon^{-1}$, independently of whether X is finite.*

Proof. Choose an $\epsilon/(6P_0)$-net of the radius-R ball with at most $(1 + 12P_0R/\epsilon)^q$ points. Apply the finite-set theorem to this net with error $\epsilon/2$. The original and compressed networks have Lipschitz constants at most P_0 and $2P_0$, respectively. Their difference therefore changes by at most $\epsilon/2$ between any point and its nearest net point. This proves uniform error at most ϵ. The logarithm of the net size is polynomial in the displayed quantities. □

Corollary 11.3 (A width-free sufficient representation condition)**.** *Suppose f has some finite-width, depth-D_0 tanh representation with unit global margin and product of Frobenius norms at most $P_0 \geq 1$. For each fixed $\zeta > 0$, there are universal polynomials and a universal constant c such that its unrestricted Gaussian posterior sample complexity is polynomial in*

$$d, D_0, P_0, L, \varepsilon^{-1}, \log(2/\delta)$$

whenever

$$L \geq D_0 + c \log(2 + d + D_0 + P_0),$$

simultaneously for every $N \geq C_L(f)$ and every $0 < \sigma^2 \leq N^{-(1+\zeta)}$. The same conclusion holds at every $L \geq 2D_0$. There is no dependence on the original width or on individual weights.

Proof. Apply Theorem 11.1 with the augmented Boolean cube, $R = \sqrt{d+1}$, $|X| = 2^d$, and approximation error $1/4$. The resulting representation has width polynomial in d, D_0, P_0, the same depth, and global signed margin at least $3/4$. Double its readout to obtain unit margin and product norm at most $4P_0$. Theorem 10.1 then applies. Its T is polynomial in d, D_0, P_0, so the required logarithmic overhead has the displayed form after increasing a universal constant. □

Together with the posterior ceiling, this identifies polynomial product norm as a substantive representation criterion for this prior. The necessity statement concerns constant-error approximations at the same learner depth; the sufficiency statement above assumes an exact Boolean representation and permits logarithmically many added layers. These qualifications prevent treating the two statements as an exact equivalence.

12 What width compression does and does not say about circuits

Corollary 12.1 (A finite-description normal form)**.** *Suppose a Boolean function on $\{-1, 1\}^d$ has an arbitrary finite-width, depth-$D \geq 2$ tanh representation with global signed margin at least one and product of Frobenius norms at most $P \geq 1$. There is a representation of the same function, with margin at least one, using at most D weighted layers, polynomial width and polynomially bounded individual weights in d, D, P, and product norm at most $8P$. Here a purely affine function of the augmented input is allowed as the case of one weighted layer. The weights may moreover be chosen dyadic with polynomial bit length, at the expense of replacing the product bound by $16P$. Consequently the function has a Boolean circuit of size polynomial in d, D, P, with no dependence on the original widths, weights, or their precision.*

Proof. Apply Theorem 11.1 on the augmented cube with error $1/4$, and double its readout. This gives width $C = \text{poly}(d, D, P)$, product at most $P_1 = 4P$, and signed margin at least $3/2$. Apply the small-prefix linearization from the proof of Theorem 10.1, with

$$\tau^2 = \frac{1}{16P_1D(d+1)^{3/2}}.$$

It changes every output by at most $1/48$. Collapse the resulting linear blocks and double the final readout. The resulting signed margin exceeds two, the product is at most $2P_1 = 8P$, and every physical weight is at most

$$R = \max\{1, 2P_1\sqrt{C}/\tau\} = \mathrm{poly}(d, D, P).$$

The number of weighted layers does not increase. If every hidden activation is removed, the remaining affine map has norm at most $2P_1$ and satisfies the same claims.

For completeness, polynomial precision can be imposed without assuming that the displayed real weights were efficiently computable. Let D' be the remaining depth and choose a polynomial $B \geq 2$ that bounds the Frobenius norm and the row absolute sum of every remaining matrix, and also bounds all input and hidden dimensions. Each matrix norm M_j satisfies

$$M_j \geq \frac{1}{\sqrt{d+1}\, B^{D'-1}}.$$

Indeed the margin gives $1 \leq |z(x)| \leq \sqrt{d+1} \prod_i M_i$, and all other norms are at most B. Round each entry to a dyadic mesh 2^{-b}, where

$$b = O\big(D' \log B + \log(d + D' + 2)\big)$$

with a sufficiently large universal constant. The matrix Frobenius perturbations are then at most $M_j/(4D')$, so the product increases by at most $(1 + 1/(4D'))^{D'} < 2$. The standard layerwise error recursion, using the row-sum bound B and the one-Lipschitz property of tanh, makes the total output change less than $1/2$ at the same precision. The initial margin exceeding two therefore leaves margin at least one. There are polynomially many entries and b is polynomial in d, D, P.

Finally, evaluate each affine map and tanh to sufficiently many binary digits and propagate a total error smaller than $1/2$. Arithmetic on these polynomial-length numbers and approximation of tanh on the polynomially bounded preactivation range use polynomial-size Boolean circuits. The output sign remains exact because of the margin. □

The converse direction is not supplied by width compression: a polynomial-size Boolean circuit is not thereby given a polynomial product-norm tanh representation. In particular, widening an existing implementation is not itself a proof of such a bound. If a polynomial product-norm implementation exists at any finite width, the corollary shows that a polynomial-width, finite-description implementation exists as well.

Nor does this normal form impose constant threshold depth. The rank-one approximation argument gives retained depth of order $P^4 \log(2P)$, after restriction to the at most P^2 relevant coordinates, which is only a polynomial depth bound when P is polynomial. Its combination with width compression therefore gives no inclusion in TC^0 or NC^1 and no separation of those circuit classes.

13 Product norm controls Boolean description size independently of depth

The rank-one bottleneck strategy of Golowich–Rakhlin–Shamir [5] underlies depth-independent norm-based complexity bounds. Here we use it for a more specific representation statement: global Boolean margin permits a short, polynomial-norm tanh implementation of the scalar suffix, without carrying a constant rail through the retained prefix. The conclusion bounds a finite Boolean description independently of the original depth as well as width.

Lemma 13.1 (Relevant coordinates and freezing)**.** *A Boolean function with a globally unit-margin tanh representation of product norm at most P depends on at most P^2 coordinates. After deleting all irrelevant coordinates, it has a representation of the same depth and no larger product norm.*

Proof. Let $T = \prod_{j\geq 2} \|W^j\|_F$. For every relevant coordinate i, some cube edge in that coordinate has opposite labels at its endpoints. The two outputs differ by at least two, whereas the Lipschitz bound along that edge is $2\|W^1 e_i\|_2 T$. Thus $\|W^1 e_i\|_2 T \geq 1$. Squaring and summing over the relevant coordinates gives their number at most $\|W^1\|_F^2 T^2 \leq P^2$.

Freeze each irrelevant coordinate i to an independent uniform sign a_i. This replaces the first-layer bias column w_0 by $w_0 + \sum_i a_i w_i$, while retaining the relevant columns. Its expected squared Frobenius norm is exactly the original squared Frobenius norm. Some assignment therefore does not increase that norm. Since the function is independent of these coordinates, the reduced network retains its unit margin on the full reduced cube. □

Lemma 13.2 (A bias-free odd scalar classifier)**.** *Let $P, R \geq 1$ and let $g : [-R, R] \to \mathbb{R}$ be odd and P-Lipschitz. Let $T \subset [-R, R]$ be finite and nonempty, with $|g(t)| \geq 7/8$ for $t \in T$. There is a tanh network with two hidden layers, a linear readout and no biases, whose sign agrees with g on T and whose signed output there is at least $7/4$. Its width, weights, and product of Frobenius norms are bounded by universal polynomials in P, R.*

Proof. Extend the set to $T \cup (-T)$ and sort its positive absolute values. All are at least $7/(8P)$, since $g(0) = 0$. Consecutive values with opposite signs of g are separated by at least $7/(4P)$. Place a threshold θ_j at the midpoint of each such transition gap. There are $m \leq 2PR$ transitions, every $0 < \theta_j \leq R$, and every actual absolute value is at distance at least $7/(8P)$ from every threshold.

The resulting odd step classifier has the exact expression

$$F(t) = c_0 \operatorname{sign}(t) + \sum_{j=1}^{m} c_j \operatorname{sign}\big(t - \theta_j \operatorname{sign}(t)\big), \qquad |c_0| \leq 1, \quad c_j \in \{-1, 1\}.$$

Indeed, the positive-side label changes by $2c_j$ at transition j. The changes alternate, so the coefficient $c_0 = \ell_0 + \sum_j c_j$ is in $\{-1, 0, 1\}$, where ℓ_0 is the first positive-side label. The displayed formula is the odd extension of this step classifier and equals $\operatorname{sign} g(t)$ on T.

Set

$$\rho = \frac{1}{2P}, \qquad \delta = \min\left\{\frac{\rho}{4R}, \frac{1}{32(m+1)}\right\}, \qquad A = \frac{1}{2\rho}\log\frac{2}{\delta},$$
$$\alpha^2 = \frac{\rho}{4R^3}, \qquad B = \rho^{-1}\log(64(m+1)), \qquad c = \operatorname{arctanh}(1/2).$$

The first hidden layer computes $u(t) = \tanh(\alpha t)$ and $s(t) = \tanh(At)$. On $T \cup (-T)$,

$$|u(t)/\alpha - t| \leq \rho/12, \qquad |s(t) - \operatorname{sign}(t)| \leq \delta.$$

Thus replacing $t - \theta_j \operatorname{sign}(t)$ by $u(t)/\alpha - \theta_j s(t)$ changes it by at most $\rho/12 + R\delta \leq \rho/3$. The original magnitude exceeds ρ, so its sign is unchanged and its new magnitude exceeds $\rho/2$. The second hidden layer computes

$$r(t) = \tanh(cs(t)), \qquad v_j(t) = \tanh\big(B[u(t)/\alpha - \theta_j s(t)]\big).$$

Consequently $2r(t)$ approximates $\operatorname{sign}(t)$ to error at most 2δ, and each v_j approximates its desired sign to error at most $1/[32(m+1)]$. Therefore

$$z(t) = 2\left[2c_0 r(t) + \sum_{j=1}^{m} c_j v_j(t)\right]$$

has signed value at least $2[1 - 2\delta - m/(32(m+1))] \geq 15/8 > 7/4$ on T. This includes $m = 0$, when only the sign rail is needed.

The three squared matrix norms are bounded respectively by

$$\alpha^2 + A^2, \qquad c^2 + mB^2(\alpha^{-2} + R^2), \qquad 16 + 4m.$$

Since $m \leq 2PR$, these expressions, their product, and every physical weight are polynomially bounded in P, R. All biases are zero. □

Theorem 13.3 (A depth-independent Boolean normal form)**.** *There is a universal polynomial p with the following property. Suppose $f : \{-1,1\}^d \to \{-1,1\}$ has some finite-depth, finite-width tanh representation with unit global margin and product of Frobenius norms at most $P \geq 1$. Then f has a globally unit-margin representation using only its relevant input coordinates, with depth, hidden widths, individual weights, and product of Frobenius norms at most $p(P)$. Its weights can be chosen dyadic with at most $p(P)$ bits each. The bounds are independent of the original depth, widths, individual weights and their precision. There are at most P^2 relevant coordinates.*

Proof. First use Lemma 13.1. If no coordinate is relevant, a constant network of width one and depth two suffices. Otherwise write $k \leq P^2$ for the number of remaining coordinates and $R = \sqrt{k+1}$. Let D be the remaining original depth, and put

$$r = \left\lceil 128 P^2 R^2 \log(2PR) \right\rceil.$$

If $D \leq r + 2$, retain the network for the moment. Otherwise we produce a representation with at most $r + 2$ weighted layers and polynomial product norm.

Write $M_j = \|W^j\|_F$, $s_j = \|W^j\|_{\mathrm{op}}$ and $\mathrm{sr}_j = M_j^2/s_j^2$. Unit margin implies $\prod_j s_j \geq 1/R$, because all activations are zero-preserving and one-Lipschitz. Hence

$$\sum_j \log \mathrm{sr}_j = 2 \log \frac{\prod_j M_j}{\prod_j s_j} \leq 2 \log(PR).$$

Among the first r hidden layers choose one, j, with $\log \mathrm{sr}_j \leq 2\log(PR)/r$. Replace W^j by its leading singular component $s_j u v^T$. The discarded operator norm is at most $M_j \sqrt{1 - 1/\mathrm{sr}_j} \leq M_j \sqrt{\log \mathrm{sr}_j}$. Propagation through the original prefix and suffix therefore changes every output by at most

$$PR\sqrt{2\log(PR)/r} \leq 1/8.$$

The modified network retains the target's signs and margin at least $7/8$.

Put $S = \prod_{i<j} M_i$, with $S = 1$ if $j = 1$, and normalize the bottleneck scalar as $t(x) = v^T h^{j-1}(x)/S$. Its absolute value is at most R. Everything from the rank-one matrix onward is an odd scalar function $g(t)$ with Lipschitz constant at most $s_j S \prod_{i>j} M_i \leq P$. Apply Lemma 13.2 on the finite set of these actual scalar values.

Attach its two-hidden-layer classifier to the original prefix. The first two new rows are $\alpha v^T/S$ and Av^T/S. Their common factor $1/S$ cancels the prefix product S in the total Frobenius product. Thus the complete new network has polynomial product norm in P, R, independent of every original dimension and weight scale. Its depth is $j + 2 \leq r + 2$, and it has the required global signs with margin at least $7/4$.

In either case we now have depth and product norm polynomial in P, while widths may still be arbitrary. Apply the finite-description normal form, Corollary 12.1, with input dimension $k \leq P^2$. It gives polynomial width, physical weights and dyadic bit length, keeping unit margin and a

polynomial product norm. Restore zero columns for the irrelevant original coordinates if a d-input network is desired. If the normal form is purely affine, apply tanh to that affine output and append the readout 2; its unit margin is preserved and all bounds remain polynomial, now within the usual architecture with a hidden layer. □

Corollary 13.4 (Counting globally margined product-norm classifiers)**.** *Let $\mathcal{H}_P$ contain all Boolean functions admitting any finite-depth, finite-width globally unit-margin tanh representation with product norm at most $P \geq 1$. For a universal polynomial p,*

$$\log |\mathcal{H}_P| \leq p(P) + P^2 \log(d+1).$$

Consequently, with probability at least $1-\delta$, every member of $\mathcal{H}_P$ consistent with n independent realizable examples has error at most ε, provided

$$n \geq \frac{p(P) + P^2 \log(d+1) + \log(1/\delta)}{\varepsilon}.$$

There is no depth or width restriction in this simultaneous event.

Proof. There are at most $(d+1)^{\lfloor P^2 \rfloor}$ possible ordered lists of relevant coordinates, allowing a padding symbol. Once the coordinates are chosen, Theorem 13.3 encodes the entire polynomial-size dyadic network in $p(P)$ bits, after enlarging a universal polynomial. Count these descriptions and apply the usual union bound: an error-ε classifier is consistent with n independent examples with probability at most $e^{-n\varepsilon}$. □

In particular, counting implies that most Boolean functions of k variables require product norm at least 2^{ck} for some universal $c > 0$, even when arbitrary depth and width are allowed. This is not a small-circuit lower bound: those generic functions also require exponentially large ordinary Boolean descriptions.

14 A shallow product-norm obstruction that depth removes

One possible route to a stronger normal form would replace every small-product deep network by a one-hidden-layer network with polynomial product cost. This is false, even for a Boolean target that the present unrestricted posterior learns with polynomial sample complexity. The obstruction concerns shallow representation, rather than the learner's ability to use a different deep representation.

For $x, y \in \{-1,1\}^n$, put $u_i = (1-x_i)/2$, $v_i = (1-y_i)/2$, and define

$$\mathrm{IP}_n(x,y) = (-1)^{\sum_{i=1}^n u_i v_i}.$$

This is the Boolean inner-product function on $2n$ input coordinates. As elsewhere, depth counts weighted layers: one hidden layer has depth two, and two hidden layers have depth three.

Theorem 14.1 (Exponential shallow cost, polynomial deep cost)**.** *For every $n \geq 1$, every finite-width one-hidden-layer tanh network*

$$z(x,y) = \sum_{a=1}^{m} v_a \tanh\bigl(w_a^T (1,x,y)\bigr)$$

with $\mathrm{IP}_n(x,y)z(x,y) \geq 1$ *on the entire cube satisfies*

$$\|v\|_2 \|W\|_F \geq \frac{2^{n/2}}{(2n+3)\sqrt{2n+1}}, \qquad W = (w_a^T)_{a=1}^m.$$

There is, however, a depth-three, width-$(n+1)$ *tanh realization in the present architecture, with signed margin at least one and*

$$\prod_{\ell=1}^{3} \|W^\ell\|_F \leq 20 n^{5/2} \log^2(8n).$$

It uses only the augmented first-layer input, with no later biases. Its individual weights are polynomially bounded in n*.*

We first prove a correlation estimate. All expectations in this section are under independent uniform Boolean inputs.

Lemma 14.2 (Halfspaces against a Hadamard sign matrix)**.** *Let* $N = 2^n$ *and* $H_{uv} = (-1)^{u \cdot v}$ *for* $u, v \in \{0,1\}^n$*. If* H' *is obtained from* H *by independently changing row and column signs, then every matrix*

$$A_{uv} = \operatorname{sign}(\alpha_u + \beta_v + c)$$

satisfies

$$\left| \frac{1}{N^2} \sum_{u,v} H'_{uv} A_{uv} \right| \leq \kappa_n := \frac{2n+3}{\sqrt{N}}.$$

The choice of sign at zero is immaterial.

Proof. The Hadamard identity $HH^T = NI$ gives $\|H'\|_{\mathrm{op}} = \sqrt{N}$. Order the columns by decreasing β_v. The positive entries of each row of A then form a prefix. Thus $A = 2B - J$, where B is the indicator of these row prefixes and J is the all-ones matrix.

Decompose each prefix into aligned dyadic intervals, using at most one interval of each length 2^ℓ, $0 \leq \ell \leq n$. At a fixed scale, the resulting part of B is a sum of rectangles $\mathbf{1}_{R_I} \mathbf{1}_I^T$, where the column intervals I are disjoint and each row belongs to at most one R_I. Its nuclear norm is at most

$$\sum_I \sqrt{|R_I| 2^\ell} \leq \sqrt{\frac{N}{2^\ell} 2^\ell \sum_I |R_I|} \leq N.$$

Summing over the $n+1$ scales and using $\|J\|_* = N$ gives

$$\|A\|_* \leq 2\|B\|_* + \|J\|_* \leq N(2n+3).$$

By nuclear/operator norm duality, the normalized inner product is at most $N^{-2}\|H'\|_{\mathrm{op}}\|A\|_* \leq \kappa_n$. Column permutations preserve the norms used in this argument. □

Proof of the lower bound in Theorem 14.1. Every affine threshold of $(1, x, y)$ has the matrix form in Lemma 14.2. The function $\phi(t) = \operatorname{sech}^2(t)$ tends to zero at both infinities and has total variation two. Hence

$$\phi(s) = \frac{1}{2} \int_{\mathbb{R}} \phi'(t) \operatorname{sign}(s-t)\, dt, \qquad \frac{1}{2} \int_{\mathbb{R}} |\phi'(t)|\, dt = 1.$$

Integrating the halfspace correlation estimate therefore shows that

$$\left| \mathbb{E} H' \operatorname{sech}^2(w^T(1,x,y)) \right| \leq \kappa_n$$

for every w and every allowed row/column sign change H'.

Write $\bar{x} = (1, x, y)$. Since tanh vanishes at zero,

$$\tanh(w^T \bar{x}) = \int_0^1 \sum_{i=0}^{2n} w_i \bar{x}_i \operatorname{sech}^2(s w^T \bar{x})\, ds.$$

Multiplying H by any coordinate $\bar{x}_i$ changes only row signs, only column signs, or neither. Consequently

$$\left|\mathbb{EIP}_n \tanh(w^T \bar{x})\right| \le \kappa_n \|w\|_1 \le \kappa_n \sqrt{2n+1}\|w\|_2. \tag{1}$$

The unit-margin hypothesis gives $\mathbb{EIP}_n z \ge 1$. Sum (1) over the hidden neurons and apply Cauchy–Schwarz:

$$1 \le \kappa_n \sqrt{2n+1} \sum_a |v_a| \|w_a\|_2 \le \kappa_n \sqrt{2n+1} \|v\|_2 \|W\|_F.$$

This proves the lower bound, with no restriction on hidden width or individual coefficient magnitudes. It also lower-bounds the shallow variation cost $\sum_a |v_a| \|w_a\|_2$ by the same quantity. □

Proof of the upper bound in Theorem 14.1. Set $\lambda = \log(8n)$ and $c = \operatorname{atanh}(1/2)$. The first hidden layer computes

$$h_i = \tanh(\lambda(-x_i - y_i - 1)), \quad 1 \le i \le n, \qquad h_0 = \tanh(c) = \tfrac{1}{2}.$$

Let $s_i = +1$ exactly when $u_i = v_i = 1$. The corresponding affine preactivation has absolute value at least one, so

$$|h_i - s_i| \le 2e^{-2\lambda} = \frac{1}{32n^2}.$$

The second hidden layer computes

$$r_j = \tanh\left(\lambda\left[\sum_{i=1}^n h_i + n - 2j + 1\right]\right), \quad 1 \le j \le n, \qquad r_0 = \tanh(2ch_0) = \tfrac{1}{2}.$$

The constant term in row j is supplied by weight $2\lambda(n - 2j + 1)$ from h_0. Put $m = \sum_i u_i v_i$. The intended second-layer sign is $q_j = \operatorname{sign}(2m - 2j + 1)$. The perturbation of this odd integer is at most $1/(32n)$, so its magnitude remains at least $3/4$. It follows that

$$|r_j - q_j| \le 2e^{-3\lambda/2} \le \frac{1}{8n}.$$

With $b_0 = 1 + \sum_{j=1}^n (-1)^j \in \{0, 1\}$, the identity

$$(-1)^m = b_0 + \sum_{j=1}^n (-1)^j q_j$$

shows that the linear readout

$$z = 2\sum_{j=1}^n (-1)^j r_j + 4b_0 r_0$$

has signed margin at least $2(1 - 1/8) = 7/4$.

The first squared Frobenius norm is $3n\lambda^2 + c^2 \le 4n\lambda^2$. For the second it is

$$n^2\lambda^2 + 4\lambda^2 \sum_{j=1}^n (n - 2j + 1)^2 + 4c^2 = n^2\lambda^2 + \frac{4n(n^2-1)}{3}\lambda^2 + 4c^2 \le 4n^3\lambda^2.$$

The readout squared norm is $4n + 16b_0^2 \le 20n$. Thus the three norms are bounded by $2\sqrt{n}\lambda$, $2n^{3/2}\lambda$, and $5\sqrt{n}$, whose product is the asserted upper bound. The construction has uniform hidden width $n + 1$ and no internal biases. □

The theorem rules out a polynomial-cost flattening into a single hidden layer, even when arbitrarily many hidden units and arbitrarily precise real coefficients are allowed. It does not rule out a polynomial-size constant-depth threshold representation; the displayed target already has one. Nor does this unit-margin exact-representation lower bound by itself lower-bound the error of approximate sign representations.

In fact Corollary 22.5 applied to the explicit depth-three construction gives the original unrestricted posterior a sample bound polynomial in $n, L, \varepsilon^{-1}, \log(2/\delta)$ for every $L \ge 3$, every actual width $N \ge C_L(\mathrm{IP}_n)$, and every $0 < \sigma^2 \le N^{-(1+\zeta)}$, for each fixed $\zeta > 0$. An exponential shallow norm is therefore compatible with efficient statistical learning by this deep posterior. It cannot on its own supply the missing unrestricted-depth counterexample.

15 Excess Gaussian energy and intrinsic product complexity

Define a function-level complexity, allowing all finite architectures, by

$$\mathfrak{P}(f) = \inf\left\{ \prod_j \|W^j\|_F : f(x) z_w(x) \ge 1 \text{ on the entire cube} \right\}.$$

The infimum permits arbitrary finite depth and width, retains tanh hidden layers and a linear readout, and allows biases only through the augmented input. It is finite, for example by a truth-table shallow construction. It is also at least one: the network extends oddly to the full augmented cube $\{-1, 1\}^{d+1}$ and has absolute margin at least one there. This nonconstant odd Boolean extension has a bichromatic cube edge. Along that edge the product of layer operator norms, and hence the Frobenius product, must be at least one.

Let $J_{L,N}(f)$ be the infimum of the Gaussian energy κ_N over width-N, depth-L unit-margin representations of f, with $J_{L,N}(f) = +\infty$ if none exists. This is a deterministic energy quantity, not a finite-temperature posterior probability.

Theorem 15.1 (Intrinsic product complexity characterizes excess energy)**.** *For every $d \ge 1$, $L \ge 2$ and $N \ge 1$,*

$$J_{L,N}(f) \ge L + \log d + 2 \log \mathfrak{P}(f).$$

Conversely, there are universal polynomials p, q and a universal constant c such that

$$L \ge p(\mathfrak{P}(f)) + c \log(d + 1), \qquad N \ge q(d, \mathfrak{P}(f))$$

imply

$$J_{L,N}(f) \le L + c \log(2d\mathfrak{P}(f)).$$

The upper-bound witness can be chosen with all physical weights at most $q(d, \mathfrak{P}(f))$. The polynomials do not depend on any supplied representation's depth, width, or coefficient precision.

Proof. The product of the L weighted layer energies is $d \prod_j \|W^j\|_F^2$. Arithmetic–geometric mean and $e^u \ge 1 + u$ therefore give

$$\kappa_N \ge L \left(d \prod_j \|W^j\|_F^2 \right)^{1/L} \ge L + \log d + 2 \log \mathfrak{P}(f).$$

This proves the lower bound without any width or depth restriction.

Choose a finite representation of product norm less than $2\mathfrak{P}(f)$. The depth-independent normal form supplies a unit-margin representation whose depth D_0, width C, physical weights and product norm P_0 are all polynomial in $\mathfrak{P}(f)$. Apply the product-target construction of Theorem 10.1. Its width threshold M and transmission parameter T are polynomial in d, C, D_0, P_0, hence polynomial in $d, \mathfrak{P}(f)$, and do not depend on the subsequently chosen learner depth. For $L \geq D_0 + \lceil \log T \rceil$, that construction has energy at most

$$L + 15 + 32 \log T.$$

If the small-prefix linearization in that proof removes every hidden activation, its separate halfspace learning conclusion is not needed here. The collapsed affine map has norm at most $2P_0$ and margin greater than one. Apply tanh to that map and append readout 2. This is a width-one shallow unit-margin representation with weights polynomial in P_0. The shallow scalar-transmission construction therefore supplies, at every sufficiently large polynomial width and every $L \geq 2 + O(\log(d + P_0))$, energy $L + O(\log(2dP_0))$. It also has polynomially bounded physical weights. Thus the affine branch has the same claimed upper-bound form.

Its physical weights are bounded by a polynomial width threshold. Since $\log T = O(\log(d+1) + \log \mathfrak{P}(f))$, increasing universal polynomials and constants gives the stated conditions and upper bound. No attainment of the infimum defining $\mathfrak{P}(f)$ was assumed. □

Corollary 15.2 (Learning from intrinsic product complexity)**.** *For each fixed $\zeta > 0$, the unrestricted Gaussian posterior has a sample bound polynomial in*

$$d, \mathfrak{P}(f), L, \varepsilon^{-1}, \log(2/\delta)$$

at every $L \geq p(\mathfrak{P}(f)) + c \log(d+1)$, simultaneously for all actual widths $N \geq C_L(f)$ and all $0 < \sigma^2 \leq N^{-(1+\zeta)}$.

Proof. Use the same depth-independent normal form and apply the all-width product-target learning theorem. Its intrinsic parameters are all polynomial in $\mathfrak{P}(f)$. Its finite-width branch covers the interval between the actual minimum width and the constructed polynomial-width witness threshold. □

Thus, once depth and width are sufficiently large, the excess above the depth baseline is, up to universal constants, the logarithm of intrinsic product complexity together with the input normalization cost $\log d$. This does not identify $\mathfrak{P}(f)$ with ordinary circuit size. The normal form proves that ordinary Boolean description size is polynomial in $\mathfrak{P}(f)$; a converse polynomial bound for arbitrary small deep circuits remains the central unresolved representation question. The learning corollary also retains its displayed learner-depth threshold.

16 Depth alignment and majority combinations

A possible route from distribution-dependent approximation to exact representation uses minimax and a majority of approximants. The unresolved step is combining their different nonlinear computations at polynomial product cost. Unequal depths alone are not an obstruction: a scalar readout can be padded at constant product cost. A stronger conclusion holds when the approximants already share their nonlinear feature map.

Lemma 16.1 (Constant-product scalar padding)**.** *A depth-D tanh network with global signed margin at least one and Frobenius product P has, for every integer $T \geq 1$, a depth-$(D+T)$ realization of the same Boolean function, with global signed margin at least one and product at most $3P/2$. No new biases are required.*

Proof. Write z for the old readout and $R = \sqrt{d+1}$. Zero preservation and one-Lipschitzness give $|z(x)| \leq PR$. Put $M = \max\{1, PR\}$ and $\eta = (M\sqrt{T})^{-1}$, and replace the readout by

$$h_1 = \tanh(\eta z), \qquad h_{s+1} = \tanh(h_s), \qquad z_{\text{new}} = \frac{3h_T}{2\eta}.$$

The hidden absolute values are at most $T^{-1/2}$. The inequality $\tanh u \geq u - u^3/3$ for $u \geq 0$ and Bernoulli's inequality imply

$$|h_T| \geq \eta|z| \left(1 - \frac{1}{3T}\right)^T \geq \frac{2}{3}\eta|z|.$$

Oddness preserves the sign. The original readout norm is multiplied by η, the next $T-1$ scalar weights equal one, and the new readout weight is $3/(2\eta)$. Their product changes by only $3/2$. □

Proposition 16.2 (A disjoint-branch product bound)**.** *Suppose m independent branches of common weighted depth D are stacked at their first layer, remain disjoint internally, and are combined linearly at the readout. Let $a_{i\ell} > 0$ be branch i's layer norms, including its voting coefficient in its final norm, and put $P_i = \prod_\ell a_{i\ell}$. The complete product satisfies*

$$P_{\text{total}} \geq \left(\sum_{i=1}^m P_i^{2/D}\right)^{D/2}.$$

In particular, averaging m nonconstant globally unit-margin branches has product at least $m^{D/2-1}$, regardless of their internal widths or neuron-cloning choices.

Proof. Each full matrix has squared Frobenius norm $\sum_i a_{i\ell}^2$. Hölder's inequality gives

$$\sum_i \prod_\ell a_{i\ell}^{2/D} \leq \prod_\ell \left(\sum_i a_{i\ell}^2\right)^{1/D}.$$

Raise this to power $D/2$. Every nonconstant unit-margin Boolean branch has product at least one, by a bichromatic cube edge. Including its averaging coefficient gives $P_i \geq 1/m$. These are the products of the actual branches after any cloning, which proves the last assertion. □

Uniform cloning of a whole layer preserves its branch product, but neuron-by-neuron cloning can reduce it. The proposition uses the resulting products and their intrinsic margin lower bounds, so it does not require the stronger, false invariance claim. It is an obstruction to a disjoint architecture, not a lower bound on every implementation of the resulting majority function.

Theorem 16.3 (Voting with a shared nonlinear feature map)**.** *Let $h(x)$ be one common tanh feature map with prefix product S. Suppose $z_i(x) = v_i^T h(x)$ satisfy $|z_i(x)| \geq 1$ everywhere, let $g_i = \operatorname{sign} z_i$, and put $P_i = S\|v_i\|_2$. If $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, and*

$$f(x) \sum_i \lambda_i g_i(x) \geq \gamma > 0 \quad \text{at every input,}$$

then

$$\mathfrak{P}(f) \le \frac{\log(4/\gamma)}{\gamma} \sum_i \lambda_i P_i.$$

A concrete finite-width realization attains at most twice the displayed bound and adds just one weighted layer relative to the networks z_i. It uses no new biases.

Proof. Set $A = \frac{1}{2}\log(4/\gamma)$. Since $|z_i| \ge 1$, $|\tanh(Az_i) - g_i| \le \gamma/2$. Therefore

$$z_{\text{new}} = \frac{2}{\gamma} \sum_i \lambda_i \tanh(Av_i^T h)$$

has global signed margin at least one. Clone its ith new hidden neuron c_i times and divide its outgoing coefficient by c_i; this leaves the function exactly unchanged. Its product is

$$\frac{2AS}{\gamma} \sqrt{\left(\sum_i c_i \|v_i\|_2^2\right)\left(\sum_i \lambda_i^2/c_i\right)}.$$

Omit zero coefficients and take $c_i = \lceil t\lambda_i/\|v_i\|_2 \rceil$. As $t \to \infty$, the square root tends to $\sum_i \lambda_i \|v_i\|_2$. This gives the intrinsic infimum bound; a sufficiently large finite t gives its factor-two version. The vectors v_i are nonzero by their margin assumption. □

For an odd unweighted majority, $\gamma = 1/m$, so the bound is $O(m \log(m+1) \max_i P_i)$. For a fixed positive voting margin it is $O(\max_i P_i)$, independently of the number of voters.

Corollary 16.4 (Weak-to-exact representation with shared features)**.** *Fix the feature map h and a budget P. Suppose that for every probability distribution μ on the cube there is a readout v_μ with global margin $|v_\mu^T h| \ge 1$, product $S\|v_\mu\|_2 \le P$, and correlation*

$$\mathbb{E}_\mu[f(x)\,\mathrm{sign}(v_\mu^T h(x))] \ge \gamma > 0.$$

Then

$$\mathfrak{P}(f) \le P\frac{\log(4/\gamma)}{\gamma}.$$

Consequently the depth-independent normal form gives a finite polynomial-size representation in this product bound.

Proof. Retain one admissible readout for every distinct Boolean sign pattern in the specified class. The class is finite because the input set is finite. The finite minimax theorem gives a probability distribution on these readouts whose signed vote is at least γ at every input. Apply Theorem 16.3. □

The shared feature map is a substantive hypothesis. It is not supplied by allowing a different arbitrary low-product network for each input distribution. General polynomial closure under majority remains open. Preserving the original shallow depth is already false: the functions $q_j = \mathrm{sign}(2\sum_i u_i v_i - 2j + 1)$ in Section 14 have shallow polynomial product on shared pairwise-AND features, while the alternating votes $(-1)^j q_j$ and the optional constant vote form an odd majority equal to IP_n. The latter has exponential shallow product, though one additional layer gives polynomial product.

17 Depth-independent input editing

A shallow computation placed before an arbitrarily deep network can modify its Boolean function without carrying extra features through the old depth. We use this observation to change finitely many labels, and to combine a network with a predicate of small Fourier variation. The product bounds depend nearly linearly on the old product, with no dependence on the old depth.

Write $\bar{x} = (1, x)$ and let $z(u)$ denote the extension of a network to arbitrary augmented inputs $u \in \mathbb{R}^{d+1}$. If its Frobenius product is P, then

$$z(-u) = -z(u), \qquad |z(u) - z(v)| \le P\|u - v\|_2.$$

The first identity uses tanh hidden activations, a linear readout, and the absence of biases after the augmented input.

Lemma 17.1 (Vector-output cloning and precomposition)**.** *For a shallow vector map*

$$T(u) = \sum_j b_j \tanh(a_j^{\mathsf{T}} u), \qquad V = \sum_j \|a_j\|_2 \|b_j\|_2,$$

finite neuron cloning gives an identical map whose two matrix Frobenius norms have product at most $2V$. If this map is placed before a network of product P, its linear output can be merged with the old first matrix. The resulting product is at most $2PV$, and only one hidden layer is added.

Proof. Cloning neuron j exactly c_j times and dividing its outgoing vector by c_j gives product

$$\left\{ \left(\sum_j c_j \|a_j\|_2^2 \right) \left(\sum_j \|b_j\|_2^2 / c_j \right) \right\}^{1/2}.$$

After omitting zero terms, choose $c_j = \lceil t\|b_j\|_2 / \|a_j\|_2 \rceil$ and let t increase. The expression tends to V, so a finite choice is within a factor two. If the map's matrices are A, B and the old first matrix is W_1, use $\|W_1 B\|_F \le \|W_1\|_F \|B\|_F$. □

Theorem 17.2 (Changing finitely many Boolean labels)**.** *Suppose $g(x)z(\bar{x}) \ge 1$ everywhere on the d-dimensional Boolean cube, $d \ge 1$, and the supplied representation has product $P \ge 1$. Let E contain $k \ge 1$ distinct inputs, and set $f = -g$ on E and $f = g$ elsewhere. There is a unit-margin representation of f with one additional hidden layer and product at most*

$$CP\left\{ d + 1 + kd\sqrt{d+1}\,\log(8Pk\sqrt{d+1}) \right\}.$$

Here C is universal. The new prefix has finite width polynomial in $d, k, \log(P+2)$. The bound is independent of the old depth and width. The same inequality holds with $P = \mathfrak{P}(g)$, after enlarging C.

Proof. Set $R = \sqrt{d+1}$, $h = 1 - 2\mathbf{1}_E$, and

$$T(x) = h(x)\bar{x} = \bar{x} - 2\sum_{e \in E} \bar{e}\, \mathbf{1}_{\{x=e\}}.$$

Oddness gives $z(T(x)) = h(x)z(\bar{x})$, so its signed margin for f is at least one, including when g is constant.

For every $e \in E$, the affine form $s_e(x) = e^{\mathsf{T}}x - d + 1$ is $+1$ at e and at most -1 elsewhere. Put

$$\delta = \frac{1}{4PkR}, \qquad \lambda = \tfrac{1}{2}\log(8PkR), \qquad t_e(x) = \tanh(\lambda s_e(x)).$$

Then $|t_e - (2\mathbf{1}_{\{x=e\}} - 1)| \le \delta$. Consequently

$$\widetilde{T}(x) = \bar{x} - \sum_{e \in E} \bar{e}\,(1 + t_e(x)), \qquad \|\widetilde{T}(x) - T(x)\|_2 \le kR\delta = \frac{1}{4P}.$$

It follows that $2z(\widetilde{T}(x))$ has signed margin at least $3/2$ for f.

This vector map has a shallow tanh realization. Write $\alpha = \operatorname{atanh}(1/2)$, $\tau = \tanh(1)$, and let $e_0, \dots, e_d$ denote the standard coordinate vectors. Use one constant neuron with incoming row αe_0^{T} and outgoing vector $2(e_0 - \sum_{e \in E} \bar{e})$; d identity neurons with incoming rows e_i^{T} and outgoing vectors e_i/τ; and the k point neurons with incoming rows $\lambda(1 - d, e^{\mathsf{T}})$ and outgoing vectors $-\bar{e}$. The identity coordinates are exact on the cube, since $\tanh(x_i) = \tau x_i$. The variation of this vector map is at most

$$V \le 2\alpha(1 + kR) + d/\tau + k\lambda dR,$$

using $\sqrt{(d-1)^2 + d} \le d$. Apply Lemma 17.1, merge the prefix, and double the old readout. This gives product at most $4PV$, as required.

For the width assertion, the explicit choice $t = \max_j \|a_j\|_2/\|b_j\|_2$ in the cloning proof gives product at most $\sqrt{2}V$: the ceiling contributes at most $t\|a_j\|_2\|b_j\|_2$ to the incoming squared norm. For the displayed nonzero rows and columns, both this t and every ratio $\|b_j\|_2/\|a_j\|_2$ are polynomial in d, k, λ. Thus the number of clones is polynomial as claimed.

For intrinsic complexity, take a finite representation of product at most $2\mathfrak{P}(g)$ and increase the universal constant. No attainment of the defining infimum is needed. □

Lemma 17.3 (Fourier input editing)**.** *Suppose $T : \{-1,1\}^d \to \mathbb{R}^r$ has vector Fourier expansion $T = \sum_S b_S \chi_S$ and variation $A = \sum_S \|b_S\|_2 > 0$. For $0 < \eta < 1$, a shallow tanh vector map approximates T uniformly in Euclidean norm to error η with product at most*

$$C(d+1)^2 A \log\left(\frac{2(d+1)\max\{A,1\}}{\eta}\right).$$

If a network of product $P \ge 1$ has signed margin one after exact composition with T, the composed Boolean function has a representation with one additional hidden layer and product at most

$$CP(d+1)^2 A \log(8(d+1)\max\{A,1\}P).$$

Proof. For $S \ne \varnothing$, write $s = |S|$ and let t be the number of negative selected input coordinates. The exact identity

$$\chi_S(x) = b_0 + \sum_{j=1}^{s} (-1)^j \operatorname{sign}(2t - 2j + 1), \qquad b_0 = 1 + \sum_{j=1}^{s} (-1)^j \in \{0,1\},$$

represents parity using affine thresholds of integer margin one. Replacing their signs by tanh at steepness $\frac{1}{2}\log(2s/\delta)$ approximates parity to error δ. Its scalar variation is at most $C(d+1)^2 \log(2(d+1)/\delta)$. A constant neuron supplies b_0 exactly, and also supplies the empty-set parity.

Take $\delta = \min\{1/2, \eta/(2\max\{A,\eta\})\}$ and multiply the outgoing coefficients of the Sth parity approximation by b_S. The vector error is at most $A\delta \le \eta$, and the sum of vector variations has the asserted bound. Lemma 17.1 supplies the matrix-product bound. For composition take $\eta = 1/(4P)$, use the old network's P-Lipschitz property, merge the prefix, and double the final readout. □

Corollary 17.4 (Multiplication by a low-variation predicate)**.** *Let h be Boolean with Fourier variation $H = \sum_S |\widehat{h}(S)|$. Then*

$$\mathfrak{P}(gh) \le C\mathfrak{P}(g)(d+1)^3 H \log\bigl(8(d+1)^2 H\mathfrak{P}(g)\bigr).$$

In particular, multiplying by any signed parity has a cost polynomial in d and nearly linear in $\mathfrak{P}(g)$, independently of the depth of a supplied representation of g. The same order of bound holds for g AND h and g OR h, with $+1$ representing true.

Proof. For multiplication use $T(x) = h(x)\bar{x}$. Each coordinate has Fourier variation H, so its vector Fourier variation is at most $(d+1)H$. Oddness of the old network and Lemma 17.3 prove the bound. A signed parity has $H = 1$.

For AND, if $g(a) = -1$ at some cube point a, use

$$T(x) = \left(1, \frac{1+h(x)}{2}x + \frac{1-h(x)}{2}a\right).$$

It equals $\bar{x}$ when $h = +1$ and $\bar{a}$ when $h = -1$, and has vector Fourier variation at most $C(d+1)(H+1)$. If g is the constant $+1$, the answer is h, which has a direct shallow representation by the parity construction in the preceding proof. The constant -1 case is immediate. OR uses a positive witness and the reversed selector. These are existence statements; no efficient procedure for finding the witness is required. □

These results do not prove closure under majority of arbitrary polynomial-product networks. A general deep predicate need not have small Fourier variation. Constructing the vector map $h(x)(1, x)$ from such a predicate with polynomial product remains an identity-augmentation problem: carrying x through all its layers by parallel coordinates incurs the existing depth cost. Likewise, finite-label editing supplies an exact interpolation witness but does not compare it with inaccurate interpolants, so it is not by itself a noisy-label generalization theorem.

Analytic and representation foundations

The following sections provide the full supporting arguments. They are included to make this paper independent of earlier drafts in this research program.

18 Unrestricted learning at bounded widths

A robustness restriction is unnecessary when the learner's width is itself polynomially bounded. The following elementary argument retains the original all-input clean margin and Gaussian conditioning conventions, and is uniform in the Gaussian scale. It also covers analog target networks at their original depth, without digital simulation.

Let $P_N = N(d+2) + (L-2)N^2$ be the parameter count, and let $\mu_{N,t}$ denote the raw Gaussian with variances $t^2/d, t^2/N, t^2/N^2$, respectively. Thus $\kappa_N = (t^2/N)\chi^2_{P_N}$ under $\mu_{N,t}$. Fix integers $1 \le C \le M$ and suppose, for some $D \ge 1$, that

$$\mu_{N,t}\{w : f(x)z_w(x) > 1/2 \text{ for every } x\} \ge \exp(-DN/t^2). \tag{2}$$

This holds for $C \le N \le M$ and $0 < t \le 1$. Only this target-mass hypothesis is needed; posterior networks may be fragile.

Theorem 18.1. *For $0 < \varepsilon, \delta < 1$, set*

$$B = 4D + 2[d + 2 + (L-2)M] \log 2 + 4 \log(2/\varepsilon),$$
$$A = \sqrt{MB}, \qquad T = d + 1 + M(L-1), \qquad q = [4T(MA)^{L-1}]^{-1}.$$

If

$$n \geq \frac{2}{\varepsilon} \left[\log M + P_M \log(5A/q) + \log(1/\delta) \right],$$

then, with probability at least $1 - \delta$ over independent examples from any input distribution, the unrestricted Gaussian posterior has Gibbs error at most ε, simultaneously for every $C \leq N \leq M$ and $0 < t \leq 1$. Here "unrestricted" retains the original conditioning on $|z_w(x)| > 1/2$ for all inputs, but imposes no internal robustness condition. The same sample event works after any number of additional correctly labeled examples. In particular, if D and M are polynomial in d, L, C, the sample bound is polynomial in $d, L, C, \varepsilon^{-1}, \log(1/\delta)$, uniformly for all $\gamma \geq 1$ with $t^2 = N^{1-\gamma}$.

Proof. For every sample, the event used to condition the raw Gaussian contains the target event in (2). The chi-square exponential moment therefore gives

$$Q_{S,N,t}\{\kappa_N > B\} \leq \exp\left(\frac{DN}{t^2} + \frac{P_N}{2} \log 2 - \frac{NB}{4t^2} \right) \leq \varepsilon/2.$$

The last inequality uses $P_N/N \leq d + 2 + (L-2)M$, $N \geq 1$, and $t \leq 1$. This bound holds for every sample, without a probabilistic qualification.

If $\kappa_N \leq B$, every physical parameter has magnitude at most A. Round each to the nearest multiple of q, using at most $5A/q$ grid values per coordinate. For the maximum coordinatewise hidden-state discrepancies, boundedness and the Lipschitz property of tanh give

$$e_1 \leq (d+1)q, \qquad e_s \leq NAe_{s-1} + Nq, \qquad |z - z_{\text{grid}}| \leq NAe_{L-1} + Nq.$$

Consequently, uniformly over Boolean inputs,

$$|z - z_{\text{grid}}| \leq q[d + 1 + M(L-1)](MA)^{L-1} = 1/4.$$

Rounding thus preserves the entire Boolean function of every posterior network with energy at most B.

The union of these grid classifiers over $C \leq N \leq M$ has logarithmic cardinality at most $\log M + P_M \log(5A/q)$. A consistency union bound gives one sample event on which every consistent grid classifier has true error at most $\varepsilon/2$. Every good-energy posterior draw has the same function as such a classifier; the remaining posterior mass is at most $\varepsilon/2$. This proves the Gibbs bound. The finite class is independent of t, and consistency only becomes more restrictive as examples are added. □

The logarithm of the grid size satisfies

$$\log(5A/q) = \log(20AT) + (L-1)\log(MA).$$

The exponentially small parameter precision costs only a polynomial in L in this logarithm. The unresolved unrestricted problem is therefore uniformity at learner widths *larger than every polynomial in the specified target complexity and depth*; this theorem does not show those larger widths actually require more samples.

Corollary 18.2 (Same-depth bounded-weight analog targets)**.** *Suppose a width-C, depth-L tanh network realizes f with $f(x)z(x) \geq 1$ on every Boolean input and weights of magnitude at most $R \geq 1$. Fix $M \geq C$ and set*

$$\begin{aligned} K &= 2[d(d+1) + (L-1)C^2]R^2, \\ a &= [4(d+1+M(L-1))(MR)^{L-1}]^{-1}, \\ D &= \max\{1, K + [d+2+(L-2)M][2+\log(1/a)]\}. \end{aligned}$$

Then (2) holds with this D. Thus Theorem 18.1 supplies a polynomial sample bound at the original depth for every $C \leq N \leq M$, with dependence on R only through $\log R$ in the final sample bound. No robustness condition on the target implementation or prior support is needed. A target with the original strict margin $> 1/2$ satisfies the displayed hypothesis after doubling its readout.

Proof. Clone the C neurons into balanced nonempty groups of sizes n_i summing to N. Copy first-layer rows and replace internal and readout weights by W_{ij}/n_j and v_i/n_i. This preserves the exact function. Since $\max_i n_i / \min_i n_i \leq 2$, its energy is at most K, and its weights remain bounded by R. A physical-coordinate perturbation of magnitude at most a changes every output by at most $1/4$, by the recurrence above.

Use isotropic coordinates $y = (\sqrt{d}\, W^1, \sqrt{N}\, W^{\text{internal}}, Nv)$, so $y \sim \mathcal{N}(0, t^2 I)$ and $\|y_*\|^2 \leq NK$ for the cloned center. The standard-normal-coordinate box $|y_j/t - y_{*,j}/t| \leq a$ lies within the physical box, because all coordinate scalings are at least one and $t \leq 1$. Its probability is at least

$$\begin{aligned} &\left(\frac{2a}{\sqrt{2\pi}}\right)^{P_N} \exp\left[-\frac{1}{2}\sum_j (|y_{*,j}|/t + a)^2\right] \\ &\qquad \geq \exp[-NK/t^2 - P_N(2 + \log(1/a))] \geq \exp(-DN/t^2). \end{aligned}$$

Here $(u+a)^2 \leq 2u^2 + 2a^2$ and $a \leq 1$ suffice. Every network in this box has target margin at least $3/4$. □

This is an elementary finite-parameter learning result, rather than a new claim about the general VC dimension of neural networks. Its role is to isolate which width regime remains open under the exact prior conventions.

19 A width-independent tanh bound without robustness

We first establish the width-independent function-class bound used to control the low-energy signal extracted from a posterior draw. The proof accounts for the fact that tanh is not positively homogeneous.

The normalized-activation and exponential-moment method is closely related to Nakhleh–Nowak's treatment of deep neural variation spaces [6], which explicitly uses $\tanh(st)/s$ and slope discretization, and to the exponential peeling argument of Golowich–Rakhlin–Shamir [5]. We give a self-contained product-Frobenius version for the present Gaussian-energy conventions; we do not claim the underlying proof technique is new.

Theorem 19.1 (Product-Frobenius bound for tanh)**.** *Let $\mathcal{F}_{L,P}$ contain all scalar-output tanh networks with $L-1$ hidden layers, arbitrary finite widths, a linear readout, and product of layer*

Frobenius norms at most P. Biases are absent except through an augmented input. On every sample $x_1, \dots, x_n$ with $\|x_i\| \le R$,

$$\widehat{\mathfrak{R}}_n(\mathcal{F}_{L,P}) \le \frac{PR}{\sqrt{n}} \left[2 + \sqrt{2(L-1)\log\bigl(3 + n(L-1)^2\bigr)} \right].$$

The bound is independent of every hidden width and of the individual layer norms apart from their product.

Proof. If $R = 0$, every sampled input and every network output on the sample is zero, so the bound is immediate. Assume $R > 0$ below. Define $\phi_s(u) = \tanh(su)/s$ for $s > 0$, $\phi_0(u) = u$, and $\phi_\infty(u) = 0$. These functions are odd, 1-Lipschitz, and zero at zero. For finite $s, t \ge 0$ and $r \ge 0$,

$$\phi_s(ru) = r\phi_{sr}(u), \qquad |\phi_s(u) - \phi_t(u)| \le u^2|s - t|, \qquad |\phi_s(u)| \le 1/s \quad (s > 0).$$

The derivative bound follows by differentiating $\phi_s(u) = u \int_0^1 \operatorname{sech}^2(suv)\, dv$.

Let $\mathcal{G}_0$ contain the identity feature map. Inductively allow arbitrary finite-dimensional features

$$g_j(x) = \phi_{s_j}(a_j^T h(x)), \quad h \in \mathcal{G}_{\ell-1}, \quad \sum_j \|a_j\|^2 \le 1, \quad s_j \in [0, \infty].$$

All these feature vectors have norm at most R on the sample. Put $B_\ell(\epsilon) = \sup_{g \in \mathcal{G}_\ell} \| \sum_i \epsilon_i g(x_i)\|$ for independent Rademacher signs. Writing $a_j = r_j u_j$ and using the scaling identity shows

$$B_\ell \le \sup_{h,\, \|u\| \le 1,\, s \in [0,\infty]} \left| \sum_i \epsilon_i \phi_s(u^T h(x_i)) \right|.$$

For any $\eta > 0$, uniformly on $[-R, R]$ the slope family admits an η/n approximation using at most $J = 3 + n^2R^2/\eta^2$ choices: truncate $s > n/\eta$ to the zero function, and round slopes below that threshold down at mesh $\eta/(nR^2)$. The error in the Rademacher sum is at most η.

We use the convex contraction principle

$$\mathbb{E}G\left(\sup_{t \in T} \sum_i \epsilon_i \psi_i(t_i) \right) \le \mathbb{E}G\left(\sup_{t \in T} \sum_i \epsilon_i t_i \right)$$

for convex nondecreasing G and 1-Lipschitz ψ_i vanishing at zero. For completeness, the single-coordinate proof compares $X_\pm = \sup_t(a_t \pm \psi(t_k))$ and $Y_\pm = \sup_t(a_t \pm t_k)$. One has $\max(Y_+, Y_-) \ge \max(X_+, X_-)$ by $|\psi(t)| \le |t|$. An approximately maximizing pair for X_+, X_- also gives $Y_+ + Y_- \ge X_+ + X_-$, by the Lipschitz property. Convex monotonicity therefore gives $G(Y_+) + G(Y_-) \ge G(X_+) + G(X_-)$; iterate over coordinates. Bounded separable classes follow by approximation from finite classes. In our application, u ranges symmetrically and ϕ_s is odd, so the absolute supremum is the same as the one-sided supremum.

Apply this principle with $G(t) = e^{\lambda t}$ to each grid slope, then bound the exponential of a maximum by the sum of the exponentials:

$$\mathbb{E}e^{\lambda B_\ell} \le e^{\lambda\eta} J\, \mathbb{E}e^{\lambda B_{\ell-1}}.$$

Furthermore, $\mathbb{E}B_0 \le R\sqrt{n}$, and bounded differences gives $\mathbb{E}e^{\lambda B_0} \le e^{\lambda R\sqrt{n} + \lambda^2 nR^2/2}$. Writing $H = L - 1$, iteration and Jensen yield

$$\mathbb{E}B_H \le H\eta + R\sqrt{n} + \frac{H \log J}{\lambda} + \frac{\lambda n R^2}{2}.$$

Take $\eta = R\sqrt{n}/H$ and optimize λ to obtain $\mathbb{E}B_H \le R\sqrt{n}[2 + \sqrt{2H\log(3+nH^2)}]$.

Finally, for a particular original network, write $M_\ell = \|W^\ell\|_F$ and $S_\ell = \prod_{j\le\ell} M_j$. Dividing its ℓth hidden vector by S_ℓ changes its recursion to $\phi_{S_\ell}((W^\ell/M_\ell)g^{\ell-1})$. Thus its normalized hidden features belong to the enlarged classes above, and its scalar output is at most P times a norm-one linear readout of $\mathcal{G}_H$. Zero-norm layers give the zero function and require no normalization. This proves the theorem. All suprema are measurable by countable approximation of finite-dimensional parameter spaces and then union over widths. □

For the Gaussian energy in our model, the product of the L weighted squared layer norms is dP^2. Consequently

$$P \le d^{-1/2}(\kappa_N/L)^{L/2} \le d^{-1/2}e^{B/2} \qquad \text{if } \kappa_N \le L + B.$$

Applying the theorem to the ramp loss $\ell(yz) = \min\{1, \max\{0, 1-2yz\}\}$ gives the following explicit bound.

Corollary 19.2 (All widths at bounded excess energy)**.** *Fix $B \ge 0$. With probability at least $1-\delta$, every sample-consistent depth-L network, at any finite width, with training margins at least $1/2$ and energy $\kappa_N \le L + B$ has sign error at most*

$$\frac{4\sqrt{2}e^{B/2}}{\sqrt{n}}\left[2 + \sqrt{2(L-1)\log(3+n(L-1)^2)}\right] + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

No internal robustness assumption is imposed.

Proof. Augmented Boolean inputs have norm $\sqrt{d+1}$, so $PR \le \sqrt{2}e^{B/2}$. The ramp loss is bounded by one and is 2-Lipschitz. Symmetrization, scalar contraction, and bounded differences give population loss at most empirical loss plus four times the network Rademacher bound and the stated confidence term. Empirical loss is zero and population loss upper-bounds sign error. □

Lemma 19.3 (A scalar transmission construction)**.** *For every real $D > 0$ and integer $m \ge 2$, there are positive scalar weights $\alpha, b_2, \dots, b_{m-1}, c$ such that*

$$r_1 = \tanh\alpha, \quad r_s = \tanh(b_s r_{s-1}), \quad c r_{m-1} = 1/2,$$

and, writing $D' = \max\{D, 4\}$,

$$D\alpha^2 + \sum_{s=2}^{m-1} b_s^2 + c^2 \le (m+12)(D'/4)^{1/m}, \qquad c^2 = (D'/4)^{1/m}.$$

Proof. Set $q = (D'/4)^{1/(2m)}$, $k_s = \min\{s, m-s\}$, and $r_s = q^s/(\sqrt{D'}k_s)$. Then $0 < r_s \le 1/(2qk_s) \le 1/2$ and $r_{m-1} = 1/(2q)$. Choose $\alpha = \operatorname{atanh} r_1$, $b_s = \operatorname{atanh}(r_s)/r_{s-1}$ and $c = q$. For $0 < u \le 1/2$ the power series gives $(\operatorname{atanh}(u)/u)^2 \le 1 + u^2$. Writing $R_s = k_{s-1}/k_s$, the reciprocal pairs $k/(k+1), (k+1)/k$ satisfy

$$\left(\frac{k+1}{k}\right)^2 + \left(\frac{k}{k+1}\right)^2 - 2 \le 4/k^2.$$

Consequently $\sum_{s=2}^{m-1} R_s^2 \le m - 2 + 7$, $R_s \le 2$, and $\sum_{s=1}^{m-1} k_s^{-2} < 4$. Using $q^{2m} = D'/4$ therefore gives

$$D\alpha^2 \le q^2 + 1/4, \qquad \sum_{s=2}^{m-1} b_s^2 \le q^2(m-2+7) + 4, \qquad c^2 = q^2.$$

Their sum is at most $(m+12)q^2$. Empty internal sums at $m = 2$ satisfy the same estimates. □

20 Unrestricted finite-variance learning for $\gamma > 1$

There is a stronger bridge than concentration of the full Frobenius energy. A typical Gaussian bulk has small operator norms when $\gamma > 1$, even when its total Frobenius energy is large. Gaussian isoperimetry shows that the posterior lies near such a bulk, with the remaining signal having energy close to a target witness's. Removing the bulk then preserves all Boolean signs under an explicit product-norm condition. No additional restriction of prior support is used.

Write $\sigma^2 = t^2/N$, so the original Gaussian has independent entry variances $N\sigma^2/d, \sigma^2, \sigma^2/N$. Its energy coordinates are

$$A_1 = \sqrt{d/N}\, W^1, \qquad A_s = W^s \ (2 \le s < L), \qquad A_L = \sqrt{N}\, v,$$

and all their entries are independent $\mathcal{N}(0, \sigma^2)$. Their squared Euclidean norm is exactly κ_N.

Theorem 20.1 (Unrestricted learning above exponent one)**.** *Fix integers $d \ge 1$, $L \ge 2$, $C \ge 1$, and a real $K \ge L$. Use tanh hidden activations and a linear readout. Suppose that for every $N \ge C$ there is a width-N, depth-L target witness satisfying*

$$f(x)z(x) \ge 1 \quad \textit{for every Boolean input}, \qquad \kappa_N \le K.$$

Put

$$Q_* = \left(\frac{K+2}{L}\right)^{L/2}.$$

For every fixed $\zeta > 0$, every $0 < \varepsilon, \delta < 1$, and every distribution D on the Boolean cube, the original Gaussian learner, conditioned only on the original global clean margin and noiseless training consistency, has a sample bound polynomial in

$$d, L, C, K, Q_*, \varepsilon^{-1}, \log(1/\delta)$$

which, with probability at least $1 - \delta$ over independent training inputs drawn from D and labeled by f, gives Gibbs error at most ε, simultaneously for all $N \ge C$ and all $0 < \sigma^2 \le N^{-(1+\zeta)}$. Thus it covers all $\gamma \ge 1 + \zeta$ at their actual finite Gaussian scales. The polynomial degree may depend on ζ; no uniform claim as $\zeta \downarrow 0$ is made.

We prove the statement with explicit width thresholds and sample inequalities. The hypothesis $K \ge L$ costs nothing: averaging over the cube and using AM–GM shows that a global unit-margin witness has $\kappa_N \ge Ld^{1/L} \ge L$.

A perturbation estimate using operator norms. Normalize hidden states by $\sqrt{N}$ and the augmented input by $\sqrt{d}$. The input norm is at most $\sqrt{2}$, all coordinate activations remain 1-Lipschitz and zero-preserving, and the layer maps are the A_s above. If U has energy at most $K + 1$, and an additive weight perturbation B has operator norm at most η in every normalized layer, telescoping one layer at a time gives

$$\sup_x |z_{U+B}(x) - z_U(x)| \le \sqrt{2}\, L\eta \left[\frac{K + 1 + 2\eta\sqrt{L(K+1)} + L\eta^2}{L-1}\right]^{(L-1)/2}. \tag{3}$$

Indeed, each summand is at most $\sqrt{2}\eta$ times a product of $L - 1$ other layer norms. The sum of squares of all perturbed layer-norm bounds is at most the numerator, and AM–GM bounds each such product. Set

$$A = \left(\frac{K+2}{L-1}\right)^{(L-1)/2}, \qquad \eta = \min\left\{1, \frac{1}{4\sqrt{L(K+1)}}, \frac{1}{8\sqrt{2}\, LA}\right\}.$$

The numerator in (3) is at most $K+2$, so the output change is at most $1/8$. Furthermore $A \le \sqrt{e}\, Q_*$, hence η^{-1} is polynomial in L, K, Q_*.

A symmetric typical Gaussian set. For completely explicit constants, put

$$a_d^2 = 2(d+1)\log 2 + 4\log 8, \qquad b_L^2 = 16\log 9 + 8\log(8L), \qquad b = \max\{a_d, b_L\}.$$

Let T consist of bulk matrices whose first and last energy-coordinate Frobenius norms are at most ta_d, and whose internal operator norms are at most tb_L. Then T is closed, convex and centrally symmetric, and

$$\mu_{N,t}(T) \ge 1/2. \tag{4}$$

For the first block, its squared Frobenius norm is $(t^2/N)\chi^2_{N(d+1)}$; the readout is the same expression with $d+1$ replaced by one. The exponential moment of χ^2 bounds each failure probability by $1/8$. For an internal matrix $tG/\sqrt{N}$, a $1/4$-net of the unit sphere has at most 9^N points, and the two-net scalar Gaussian bound gives

$$\Pr\{\|tG/\sqrt{N}\|_{\mathrm{op}} > tb_L\} \le 2\exp(2N\log 9 - Nb_L^2/8) \le \frac{1}{4L}.$$

Union bounding over the $L-2$ internal blocks proves (4). Every normalized block of a member of T has operator norm at most tb.

Target mass by a Gaussian shift. Suppose $tb \le \eta$. For a target witness w_*, the shifted set $w_* + T$ has target margin at least $7/8$, by (3). Central symmetry and the Gaussian change-of-mean identity yield

$$\begin{aligned}\mu(w_* + T) &= e^{-\kappa_N(w_*)/(2\sigma^2)} \int_T e^{-\langle w_*, B\rangle_\kappa/\sigma^2}\, d\mu(B) \\ &\ge e^{-K/(2\sigma^2)}\mu(T) \ge \tfrac{1}{2}e^{-K/(2\sigma^2)}.\end{aligned}$$

The inequality follows by pairing B with $-B$ and using $\cosh u \ge 1$. This lower bound holds for the raw probability of the conditioning event for every correctly labeled sample.

The posterior is close to the typical set. Let $d_\kappa(w, T)$ be Euclidean distance in energy coordinates. Gaussian isoperimetry [1] and (4) imply

$$\mu\{d_\kappa(w,T) > r\} \le e^{-r^2/(2\sigma^2)}.$$

Dividing by the preceding target-mass lower bound gives, for every sample,

$$Q_s\{d_\kappa(w,T)^2 > K+1\} \le 2e^{-1/(2\sigma^2)}. \tag{5}$$

On the complement, let B be the metric projection of w onto T and write $U = w - B$. The projection is well defined and continuous, since T is nonempty, closed and convex. Then $\kappa_N(U) \le K+1$ and $\sup_x |z_w(x) - z_U(x)| \le 1/8$. The original posterior has clean margin $> 1/2$ everywhere, so U has *exactly the same Boolean function* as w and training margins $> 3/8$. It is enough to use a margin-$1/4$ ramp loss below.

A polynomial width split. Take $\beta = \varepsilon/2$ and

$$M = \left\lceil \max\left\{C, 2, (b/\eta)^{2/\zeta}, 2\log(2/\beta)\right\}\right\rceil.$$

For every $N \geq M$ and $\sigma^2 \leq N^{-(1+\zeta)}$, one has $t \leq N^{-\zeta/2}$, hence $tb \leq \eta$, and $2e^{-1/(2\sigma^2)} \leq \beta$. The product norm of U is at most $d^{-1/2}[(K+1)/L]^{L/2}$. Write

$$G = \sqrt{(d+1)/d}\left(\frac{K+1}{L}\right)^{L/2}, \qquad H = L - 1.$$

The product-norm Rademacher bound, now with the 4-Lipschitz ramp loss $\min\{1, \max\{0, 1-4yz\}\}$, gives one sample event of probability at least $1-\delta/2$ on which every such consistent residual network has error at most

$$F_n = \frac{8G}{\sqrt{n}}\left[2 + \sqrt{2H\log(3+nH^2)}\right] + \sqrt{\frac{\log(2/\delta)}{2n}}. \tag{6}$$

This event already includes all finite widths. When $F_n \leq \varepsilon/2$, adding the posterior exceptional mass β proves the required bound for every $N \geq M$ and every stated scale.

For $C \leq N \leq M$, use Theorem 18.1. Here is a target-mass constant requiring only the present witness assumption. Every witness weight is at most $R = \sqrt{MK}$. Put

$$a = [4(d+1+M(L-1))(MR)^{L-1}]^{-1},$$
$$D_M = \max\{1, K + [d+2+(L-2)M][2+\log(1/a)]\}.$$

The same Gaussian-coordinate box calculation as in the finite-width proof has physical perturbation at most a, retains target margin at least $3/4$, and has mass at least $\exp(-ND_M/t^2)$. Applying that theorem with error ε and failure probability $\delta/2$ controls all these widths and all $t \leq 1$ simultaneously. Its finite-grid logarithm is polynomial in $M, d, L, \log K$. The union of the two sample events proves the theorem. Finally, $G \leq \sqrt{2}Q_*$, $b^2 = O(d + \log L)$ and $A \leq \sqrt{e}Q_*$ show that M and a sample size satisfying all the displayed inequalities are polynomial in the stated arguments for fixed ζ. This proves the asserted quantitative dependence.

Scope. This argument does not apply at $\gamma = 1$: the Gaussian bulk's operator norm does not tend to zero with width there. It also does not make Q_* polynomial for every small circuit. If a fixed base witness can be padded with energy $K = L + B$, where B is independent of the added depth, then $Q_* \leq e^{(B+2)/2}$. The theorem therefore removes exponential dependence on *additional depth*, at every width and every fixed exponent above one, while retaining a potentially exponential dependence on base-witness energy. Stronger target-specific energy estimates can make Q_* itself polynomial.

21 A polynomially quantified mean-field prefactor window

The bulk-subtraction proof depends on the prefactor t, not directly on the exponent γ. This gives a separate finite-temperature statement at mean-field scaling. It does not include the original critical convention $t = 1$.

Theorem 21.1 (Small prefactor at every width)**.** *Assume the witness hypothesis of Theorem 20.1, with parameters d, L, C, K and $Q_* = [(K+2)/L]^{L/2}$. Define*

$$A = \left(\frac{K+2}{L-1}\right)^{(L-1)/2}, \qquad \eta = \min\left\{1, \frac{1}{4\sqrt{L(K+1)}}, \frac{1}{8\sqrt{2}\,LA}\right\},$$

$$b^2 = \max\{2(d+1)\log 2 + 4\log 8,\ 16\log 9 + 8\log(8L)\}, \qquad t_0 = \min\left\{1, \frac{\eta}{b}, \frac{1}{\sqrt{2\log(4/\varepsilon)}}\right\}.$$

There is a sample bound polynomial in $d, L, K, Q_*, \varepsilon^{-1}, \log(2/\delta)$ such that, with probability at least $1-\delta$, the unrestricted posterior has Gibbs error at most ε, simultaneously for all $N \geq C$ and all

$$0 < t \leq t_0, \qquad \sigma^2 = t^2/N.$$

The inverse threshold t_0^{-1} is polynomial in $d, L, K, Q_*, \varepsilon^{-1}$. Neither bound depends on the realized training sample or on its size through a circular condition.

Proof. Use the typical set T and the perturbation estimate from Section 20. For every width in the statement, $tb \leq \eta$, so the target shift has raw mass at least $\frac{1}{2}e^{-K/(2\sigma^2)}$. The posterior distance tail is therefore

$$Q_S\{d_\kappa(w,T)^2 > K+1\} \leq 2e^{-1/(2\sigma^2)} = 2e^{-N/(2t^2)} \leq \varepsilon/2.$$

Every remaining draw has exactly the same Boolean function as a residual of energy at most $K+1$, with training margin greater than $3/8$. The single width-independent generalization event underlying (6) controls all of these residuals simultaneously. Choosing its right-hand side at most $\varepsilon/2$ proves the claim. This sample choice is polynomial: for example, if $G = \sqrt{(d+1)/d}[(K+1)/L]^{L/2}$, then a sufficiently large universal constant times

$$[1 + G + L + \varepsilon^{-1} + \log(4/\delta)]^{12}$$

suffices. Finally $G \leq \sqrt{2}Q_*$ and $A \leq \sqrt{e}Q_*$ give all the stated polynomial dependences. This proof requires no width split, because the small-bulk condition has been imposed directly on t. □

A sample bound without an additional depth factor. The global-margin counting theorem gives a sharper choice of sample size in Theorem 21.1. Put

$$\overline{P} = \max\left\{1, \frac{8}{3\sqrt{d}}\left(\frac{K+1}{L}\right)^{L/2}\right\}.$$

Rescale the residual readout by $8/3$. Its Boolean function now belongs to $\mathcal{H}_{\overline{P}}$ from Corollary 13.4. Therefore it suffices to use

$$n \geq \frac{2}{\varepsilon}\left[p(\overline{P}) + \overline{P}^2\log(d+1) + \log(2/\delta)\right],$$

for a universal polynomial p. In particular, if $K \leq L + B$ with B independent of the learner depth, then

$$\overline{P} \leq \max\{1, (8/3)d^{-1/2}e^{(B+1)/2}\}.$$

The required sample size has no remaining learner-depth dependence. The sufficient prefactor t_0 still depends on L, and the original critical choice $t = 1$ is still excluded. Likewise, when this argument is combined with the small-width fallback below, that finite-width part retains its polynomial dependence on L.

Corollary 21.2 (Polynomial product targets at mean-field scaling)**.** *Suppose f has an arbitrary finite-width, depth-D_0 tanh representation with unit global margin and product norm at most $P \geq 1$. There are universal polynomials p, q and a universal constant c such that, whenever*

$$L \geq D_0 + c\log(2 + d + D_0 + P) \quad \text{or} \quad L \geq 2D_0,$$

sample size at least

$$p(d, D_0, P, L, \varepsilon^{-1}, \log(2/\delta))$$

gives Gibbs error at most ε with probability at least $1-\delta$, simultaneously for every actual width $N \geq C_L(f)$ and every mean-field scale

$$\sigma^2 = t^2/N, \qquad 0 < t \leq q(d, D_0, P, L, \varepsilon^{-1}, \log(2/\delta))^{-1}.$$

For a bounded shallow target, or for a Boolean halfspace using Theorem 4.3, the corresponding statement holds at every $L \geq 2$, with the same polynomial dependence as in those positive theorems. The original critical prior $t = 1$ is not asserted to satisfy this condition.

Proof. The width-compression and product-target constructions provide a polynomial threshold M and, at all $N \geq M$, unit-margin witnesses with K and Q_* polynomial in d, D_0, P, L. Apply Theorem 21.1 to these widths. The physical weight bound in that construction gives $C_L(f) \leq M$.

For the remaining interval $C_L(f) \leq N < M$, choose an actual minimum-width bounded-weight witness and double its readout. It now has unit margin and weights at most $2M$. Balanced cloning gives unit-margin witnesses throughout this finite width interval with energy polynomial in d, L, M. The finite-width theorem, with its Gaussian-coordinate-box denominator, gives a polynomial sample bound uniformly for all $t \leq 1$. Its constants can be bounded using M and $2M$, independently of the actual minimum width. Union bound this sample event and the large-width event, each with failure probability $\delta/2$. Their thresholds are deterministic polynomials of the displayed parameters. The same proof applies to the bounded shallow and halfspace witnesses at all $L \geq 2$. □

22 Polynomial unrestricted learning of shallow targets at every depth

A shallow target can have a small first-layer Gaussian energy at a polynomial base width simply by adding zero neurons. Its readout energy is then polynomial. Distributing that readout cost over the learner's remaining layers makes the product-norm parameter polynomial uniformly in depth. This gives a family for which neither arbitrary width nor arbitrary added depth requires a robustness restriction.

Theorem 22.1 (Shallow targets, all depths and all admissible widths)**.** *Suppose a Boolean target $f : \{-1,1\}^d \to \{-1,1\}$ has a one-hidden-layer representation*

$$z(x) = \sum_{j=1}^{C} v_j \tanh(a_j^T \bar{x}), \qquad \bar{x} = (1, x), \qquad f(x)z(x) \geq 1$$

with every coordinate of a_j and every v_j bounded in absolute value by $R \geq 1$. Let $C_L(f)$ denote its bounded-weight network complexity at depth L: the least width N admitting a realization with all weights at most N in absolute value and signed output greater than $1/2$ everywhere.

For each fixed $\zeta > 0$, the original unrestricted Gaussian learner has sample complexity polynomial in

$$d, C, R, L, \varepsilon^{-1}, \log(1/\delta),$$

for every specified learner depth $L \geq 2$ and every input distribution, simultaneously for every width $N \geq C_L(f)$ and every $\gamma \geq 1+\zeta$. With probability at least $1-\delta$ over a sample of that size, posterior Gibbs error is at most ε simultaneously at all these widths and all prior scales $0 < \sigma^2 \leq N^{-(1+\zeta)}$. The polynomial's degree may depend on ζ. The prior is conditioned only on the original global clean margin and the training labels.

Proof. Let A have rows a_j^T, and set

$$M = \left\lceil \max\{4, C, 2R, d\|A\|_F^2\} \right\rceil, \qquad T = \max\{1, M\|v\|_2^2\}.$$

Both are polynomial in d, C, R. Add zero hidden neurons to obtain width M. This preserves the function, and its first-layer and readout energies satisfy

$$E_{\text{pre}} = \frac{d}{M}\|A\|_F^2 \le 1, \qquad c^2 = M\|v\|_2^2 \le T.$$

In fact $M\|v\|_2^2 \ge 1$, since unit margin and $|\tanh| \le 1$ imply $\|v\|_2 \ge 1/\sqrt{C}$, but the maximum makes this unneeded.

For any $N \ge M$, balanced cloning of this shallow network preserves its output exactly and gives

$$E_{\text{pre}} \le 2, \qquad c^2 = N\|v_N\|_2^2 \le 2T.$$

Explicitly, assign each of the M original neurons either $\lfloor N/M \rfloor$ or $\lceil N/M \rceil$ clones, copy its incoming weights, and divide its readout weight by its number of clones. The first cost increases by at most $M\lceil N/M \rceil/N \le 2$, and the readout cost by at most $N/(M\lfloor N/M \rfloor) \le 2$. For learner depth $L = 2$ we already have a unit-margin witness with

$$K_2 = 2 + 2T, \qquad Q_*^2 = \left(\frac{K_2 + 2}{2}\right)^2 = (T+2)^2.$$

For $L \ge 3$, apply Lemma 19.3 with $D = 4c^2$ and $m = L - 1$. Replace the old readout by a layer whose every row is $2\alpha v_N^T$. Follow it with the scalar transmission layers, implemented by equal coordinates with dense matrix entries b_s/N, and final readout entries q/N, where

$$q^2 = \max\{c^2, 1\}^{1/(L-1)} \le (2T)^{1/(L-1)}.$$

The target had margin at least one, hence in particular at least $1/2$, so the scalar construction gives margin at least $1/2$. Double its final readout to restore margin at least one. This adds $3q^2$ to the energy. Thus every width $N \ge M$ has a depth-L unit-margin witness with

$$\kappa_N \le K_L := 2 + (L + 14)(2T)^{1/(L-1)}.$$

Writing $q_*^2 = (2T)^{1/(L-1)} \ge 1$, we have $K_L + 2 \le (L + 18)q_*^2$. Consequently

$$Q_*^2 = \left(\frac{K_L + 2}{L}\right)^L \le e^{18}(2T)^{L/(L-1)} \le e^{18}(2T)^{3/2} \qquad (L \ge 3). \tag{7}$$

The exponent $L/(L-1)$ is at most $3/2$ for $L \ge 3$. Theorem 20.1, with base width M, therefore provides a sample bound polynomial in the stated quantities for all $N \ge M$.

We verify admissible smaller widths rather than silently identifying M with the target's minimum complexity. First, $C_L(f) \le M$ for all $L \ge 2$. At width M, the original incoming weights have magnitude at most $R \le M/2$. In the added scalar channel, $\alpha \le \operatorname{atanh}(1/2) < 0.55$, so its first weights $2\alpha v_j$ have magnitude at most $1.1R < M$. The scalar construction has $b_s \le 2.2q$ and final weights $2q/M$. Since $T \le MCR^2 \le M^4$ and $L \ge 3$,

$$q \le (2T)^{1/4} \le 2^{1/4}M, \qquad b_s/M < 3 \le M, \qquad 2q/M < 3 \le M.$$

Thus this depth-L, width-M circuit satisfies the bounded-weight convention and has margin at least one. The case $L = 2$ is immediate.

For $C_L(f) \le N < M$, start from a defining width-$C_L(f)$ circuit, clone it to width N, and double its readout. It has margin greater than one, all weights at most $2M$, and Gaussian energy bounded by a polynomial in d, L, M. The Gaussian box estimate and Theorem 18.1 consequently give a sample bound polynomial in $d, L, M, \varepsilon^{-1}, \log(1/\delta)$, simultaneously for all these widths and all $t \le 1$. Use failure probability $\delta/2$ for this event and $\delta/2$ for the preceding event for $N \ge M$. Their intersection proves the claim. □

The complexity parameter controlling this theorem's sample size is the size and weight bound of a *shallow* representation. Its width admissibility is the actual depth-L complexity $C_L(f)$. The theorem does not identify shallow complexity with the minimum complexity of an arbitrary deep circuit, and therefore does not settle the unrestricted problem for all small deep circuits. It also makes no claim at $\gamma = 1$.

Corollary 22.2 (Parity at arbitrary depth and admissible width)**.** *For every fixed $\zeta > 0$, parity of any $k \ge 1$ chosen coordinates in $\{-1, 1\}^d$ has unrestricted Gaussian posterior sample complexity polynomial in $d, k, L, \varepsilon^{-1}, \log(1/\delta)$, for every $L \ge 2$, every $N \ge C_L(f)$, and every $\gamma \ge 1 + \zeta$. The bound holds for every input distribution and for the original finite-variance posterior.*

Proof. Let $q = (k - \sum_{i \in S} x_i)/2 \in \{0, \dots, k\}$ count negative coordinates, and put $a = \log(16k)$. Define

$$\tau_j(x) = \tanh(a(q - j + 1/2)), \qquad 1 \le j \le k.$$

Every argument has magnitude at least $a/2$, so $|\tau_j - \operatorname{sign}(q - j + 1/2)| \le 2e^{-a} = 1/(8k)$. Writing $c_k = 1$ for even k and $c_k = 0$ for odd k, the exact step identity is

$$(-1)^q = c_k + \sum_{j=1}^{k} (-1)^j \operatorname{sign}(q - j + 1/2).$$

Hence $F = c_k + \sum_j (-1)^j \tau_j$ satisfies $|F - (-1)^q| \le 1/8$. Represent the constant by one extra hidden neuron with augmented-input weight $\operatorname{atanh}(1/2)$ and no other incoming weights. A linear readout equal to $2F$ has signed margin at least $7/4$. Its threshold-neuron readout weights are $2(-1)^j$, and its constant-neuron weight is $4c_k$. Its incoming weights have magnitude at most $a(k + 1)$. Thus $C = k + 1$ and $R = \max\{4, a(k + 1)\}$ are polynomial in k. Apply Theorem 22.1. □

Corollary 22.3 (Symmetric Boolean targets)**.** *The conclusion of Corollary 22.2 holds for any Boolean function of the Hamming weight of k selected coordinates.*

Proof. If its values are $h(0), \dots, h(k) \in \{-1, 1\}$, then

$$h(q) = \frac{h(0) + h(k)}{2} + \sum_{j=1}^{k} \frac{h(j) - h(j-1)}{2} \operatorname{sign}(q - j + 1/2).$$

Each coefficient has magnitude at most one. Replacing the signs by the same τ_j and doubling the output gives margin at least $7/4$, using the same width and weight bounds as in the parity proof. □

Corollary 22.4 (Polynomial-size DNF and CNF targets)**.** *A Boolean function on d bits represented by a DNF or CNF with $s \ge 1$ clauses has unrestricted Gaussian posterior sample complexity polynomial in $d, s, L, \varepsilon^{-1}, \log(1/\delta)$, for every specified $L \ge 2$, every $N \ge C_L(f)$, and every $\gamma \ge 1 + \zeta$, with $\zeta > 0$ fixed.*

Proof. For a DNF, let $q_j(x)$ be the number of false literals in clause j, and put $a = \log(16s)$ and $\tau_j(x) = \tanh(a(1/2 - q_j(x)))$. The exact clause indicator is $[1 + \operatorname{sign}(1/2 - q_j)]/2$. Thus

$$H(x) = s - 1 + \sum_{j=1}^{s} \operatorname{sign}(1/2 - q_j(x))$$

is -1 when no clause is satisfied and at least 1 otherwise. Replacing the signs by τ_j changes H by at most $1/8$. Doubling the resulting expression gives global signed margin at least $7/4$. It is a shallow tanh network with $s + 1$ features: the extra feature is the constant $1/2$, with readout weight $4(s - 1)$. After removing duplicate literals and discarding contradictory clauses, each clause has at most d literals. Each q_j is affine in the bits, so the remaining incoming and readout weights have magnitude at most $R = \max\{4s, a(d + 1)\}$. The shallow-target theorem applies. A CNF is the negation of a DNF in negated literals, and negating the readout preserves every bound. Constant functions are immediate. □

22.1 Targets of any fixed intrinsic depth

The shallow-target argument extends to any fixed target depth by using different numbers of clones at its successive hidden layers. The degree of the resulting polynomial may depend on the target depth, but not on the learner's additional depth.

Corollary 22.5 (Fixed target depth, arbitrary learner depth)**.** *Fix an integer $D_0 \geq 2$ and $\zeta > 0$. Suppose f has a width-C, depth-D_0 tanh realization in the present architecture, with signed output at least one on the Boolean cube and all weights bounded by $R \geq 1$. For every specified learner depth $L \geq D_0$, the original unrestricted Gaussian learner has sample complexity polynomial in*

$$d, C, R, L, \varepsilon^{-1}, \log(1/\delta).$$

The degree and constants may depend on D_0 and ζ. For every input distribution, one sample event of probability at least $1 - \delta$ gives Gibbs error at most ε simultaneously for every $N \geq C_L(f)$ and every $0 < \sigma^2 \leq N^{-(1+\zeta)}$. Here $C_L(f)$ is the actual bounded-weight network complexity defined in Theorem 22.1. The prior retains the original global clean-margin conditioning and sign-only training likelihood.

Proof. Put $H = D_0 - 1$. Pad the target's hidden layers to width C with zero neurons if necessary, and write $W^1, \ldots, W^H, v$ for its weights. Choose an integer

$$B = \left\lceil \max\left\{4, 2R, \frac{d}{C}\|W^1\|_F^2, \|W^2\|_F^2, \ldots, \|W^H\|_F^2\right\}\right\rceil, \qquad M = CB^H,$$

omitting the internal-layer terms when $H = 1$. Since $\|W^1\|_F^2 \leq C(d + 1)R^2$ and $\|W^s\|_F^2 \leq C^2R^2$ internally, B is polynomial in d, C, R, and so is M for fixed D_0.

At hidden layer s, give each original neuron exactly

$$n_s = B^{H-s}$$

active copies, and fill the other positions up to width M with zero neurons. Copy first-layer rows. For $s \geq 2$, give every edge from a copy of source j to a copy of target i weight W^s_{ij}/n_{s-1}. All unused edges are zero. Since $n_H = 1$, retain the original readout on the final active neurons. This preserves the function exactly. The first-layer and internal energy terms are respectively

$$\frac{d}{M} n_1 \|W^1\|_F^2 = \frac{d}{CB}\|W^1\|_F^2 \leq 1, \qquad \frac{n_s}{n_{s-1}}\|W^s\|_F^2 = \frac{1}{B}\|W^s\|_F^2 \leq 1.$$

Thus the energy before the readout is at most H. Put

$$T = \max\{1, M\|v\|_2^2\} \le MCR^2.$$

It is polynomial in d, C, R for fixed D_0.

Balanced cloning of this width-M network into any $N \ge M$ preserves the function and gives

$$E_{\mathrm{pre}} \le 2H, \qquad c^2 = N\|v_N\|_2^2 \le 2T.$$

If $L = D_0$, these are already unit-margin witnesses with $K = 2H + 2T$, and

$$Q_*^2 = \left(\frac{K+2}{D_0}\right)^{D_0} \le (4T)^{D_0}.$$

If $L > D_0$, set $m = L - H \ge 2$. Replace the old readout by the scalar transmission from Lemma 19.3, using $D = 4c^2$, and double the final scalar readout to obtain unit margin. Its readout cost before doubling is $q^2 = \max\{c^2, 1\}^{1/m} \le (2T)^{1/m}$. The total energy is at most

$$K_L = 2H + (m+15)(2T)^{1/m}.$$

This is the desired depth: the H old hidden layers and $m-1$ new hidden layers total $L-1$. Since $(2T)^{1/m} \ge 1$,

$$\begin{aligned} Q_*^2 &= \left(\frac{K_L+2}{L}\right)^L \\ &\le \left(1 + \frac{H+17}{L}\right)^L (2T)^{L/(L-H)} \\ &\le e^{H+17}(2T)^{(D_0+1)/2}. \end{aligned}$$

The last step uses $L/(L-H) = 1 + H/m \le 1 + H/2$. For $L \ge 2D_0$, the sharper upper bound $Q_*^2 \le e^{H+17}(2T)^2$ also holds. Both K_L and Q_* are polynomial in the stated parameters when D_0 is fixed, uniformly over the learner depth. Theorem 20.1 now controls every $N \ge M$.

It remains to check admissible widths below M. First $C_L(f) \le M$ at every $L \ge D_0$. The tapered construction has all original weights at most $R \le M/2$. In a padded scalar channel, $\alpha \le \operatorname{atanh}(1/2) < 0.55$, so its first added weights $2\alpha v_j$ have magnitude less than M. The scalar construction has $b_s < 2.2q$, and

$$T \le MCR^2 \le M^4, \qquad q \le (2T)^{1/4} \le 2^{1/4}M.$$

Its dense internal entries b_s/M and final readout entries $2q/M$ are therefore less than $3 \le M$. This gives a bounded-weight width-M witness with unit margin at each learner depth; for $L = D_0$ no scalar channel is needed.

For $C_L(f) \le N < M$, clone a defining width-$C_L(f)$ realization to width N and double its readout. Its margin is greater than one and all weights are at most $2M$. The Gaussian box estimate and Theorem 18.1 give a bound polynomial in $d, L, M, \varepsilon^{-1}, \log(1/\delta)$ for all these widths and all $t \le 1$. Allocate failure probability $\delta/2$ to this event and to the preceding event for $N \ge M$. Their intersection proves the claim. □

Thus polynomial dependence on learner depth holds for targets of any fixed intrinsic depth with polynomial target-size dependence. The width $M = CB^{D_0-1}$ and the exponents above explain why this does not give a single polynomial jointly in the intrinsic target depth and target size. The mean-field exponent $\gamma = 1$ is also still outside this result.

Corollary 22.6 (Quantitative dependence on intrinsic depth)**.** *Fix $\zeta > 0$. There are constants $a_\zeta, b_\zeta > 0$, independent of the intrinsic depth, with the following property. Suppose f has a width-C, depth-D_0 tanh realization with unit signed margin and weights bounded by $R \geq 1$, where $D_0 \geq 2$, and set $S = 2 + d + C + R$. For every specified learner depth $L \geq D_0$, the sample bound*

$$n \geq \left\lceil S^{a_\zeta D_0} \left(L + \varepsilon^{-1} + \log(2/\delta)\right)^{b_\zeta} \right\rceil$$

suffices for posterior Gibbs error at most ε, with probability at least $1 - \delta$ for every input distribution, simultaneously for all $N \geq C_L(f)$ and $0 < \sigma^2 \leq N^{-(1+\zeta)}$. The prior and likelihood are those of Corollary 22.5.

Thus the dependence on intrinsic depth is $\exp[O_\zeta(D_0 \log S)]$, while the dependence on learner depth remains polynomial with degree independent of D_0. In particular, if $D_0 \leq c \log S$ for a fixed constant c, the target-parameter factor is at most $\exp[O_{\zeta,c}((\log S)^2)]$. This is a quasipolynomial bound in S, not a single polynomial jointly in intrinsic depth and target size.

Proof. First suppose $L \geq 2D_0$. Use the tapered construction in the preceding proof, with $H = D_0 - 1$, $M = CB^H$, and $T = \max\{1, M\|v\|_2^2\}$. Its explicit definition gives $B \leq S^5$, so $M \leq S^{5D_0}$ and $T \leq MCR^2 \leq S^{7D_0}$. Put $m = L - H$. Since $L \geq 2D_0$, one has $L/m \leq 2$ and

$$K_L = 2H + (m + 15)(2T)^{1/m} = O(LT), \qquad Q_*^2 \leq e^{H+17}(2T)^{L/m} \leq e^{H+17}(2T)^2 = S^{O(D_0)}.$$

For the remaining range $D_0 \leq L < 2D_0$, do not taper the target. Instead pad its original realization to $M_0 = \lceil \max\{4, C, 2R\} \rceil \leq S^2$, and put $T_0 = \max\{1, M_0\|v\|_2^2\} \leq S^5$. Balanced cloning to every $N \geq M_0$ gives

$$E_{\mathrm{pre}} \leq 2HS^4, \qquad c^2 \leq 2T_0.$$

At $L = D_0$, retain the readout. For $L > D_0$, append the same scalar transmission with $m = L - H \geq 2$. These constructions give, respectively,

$$K_L \leq 2HS^4 + 2T_0, \qquad K_L \leq 2HS^4 + (m + 15)(2T_0)^{1/m}.$$

In either case $K_L \leq 19LS^5$. Therefore

$$Q_*^2 \leq (20S^5)^L \leq S^{7L} \leq S^{14D_0}.$$

The width-M_0 witness has weights bounded by M_0: the original weights satisfy $R \leq M_0/2$, and the scalar-channel check from the preceding proof applies because $T_0 \leq M_0^4$. Thus $C_L(f) \leq M_0$ in this range.

In both depth ranges, the polynomial in Theorem 20.1 has degree and constants depending only on ζ. Substituting the corresponding bounds gives the stated sample form above the respective base width M or M_0. For smaller admissible widths, the defining-witness and Gaussian-box argument in the preceding proof gives a sample bound polynomial in $d, L, M, \varepsilon^{-1}, \log(1/\delta)$ (or with M_0 in place of M), with degree independent of D_0. Substitution gives the same sample form. Split the failure probability between the two width ranges and enlarge a_ζ, b_ζ to finish. □

23 A finite-sample extension to targets close to a shallow function

The denominator witness need only interpolate the observed labels. This permits a limited positive extension beyond targets having an exact cheap shallow representation. The restriction below is quantitative: the target disagreement probability times the chosen sample size must fit a prescribed patch budget. It is not an unrestricted theorem for every fixed nonzero corruption rate at arbitrarily large sample sizes or confidence levels.

Lemma 23.1 (Patching finitely many Boolean inputs)**.** *Suppose a shallow width-C tanh network z_g has weights bounded by $R \geq 1$ and satisfies $g(x)z_g(x) \geq 1$ on the Boolean cube. For an integer $b \geq 0$, define*

$$Z = CR, \quad a = \log(16(b+1)(Z+1)), \quad C_b = C + 2b + 1, \qquad R_b = \max\{2R, 4Z, (d+1)a\}.$$

For any set P of at most b cube inputs, the Boolean function obtained by flipping g exactly on P has a shallow width-C_b, weight-bound-R_b representation with unit signed margin.

Proof. For each $p \in P$, let

$$h_p(x) = \tanh(a(p \cdot x - d + 1/2)), \qquad I_p(x) = \frac{1 + h_p(x)}{2}, \qquad c_p = -2z_g(p).$$

Then $|I_p(x) - \mathbf{1}\{x = p\}| \leq e^{-a}$ and $|c_p| \leq 2Z$. The ideal expression $z_g(x) + \sum_p c_p \mathbf{1}\{x = p\}$ has signed margin at least one for the flipped function. Replacing each indicator by I_p changes it by at most $2Zbe^{-a} < 1/8$. Double the resulting expression. Its original-feature readout weights are $2v_j$; its isolator readout weights are c_p; and its additional constant output is $\sum_p c_p$. Implement this constant with $|P| + 1$ neurons of activation $1/2$, each having readout weight $2\sum_p c_p/(|P| + 1)$, of magnitude at most $4Z$. Their bias is $\operatorname{atanh}(1/2)$ and their other incoming weights are zero. The number and weight bounds follow, and the signed margin is greater than $7/4$. □

Theorem 23.2 (A finite-sample target-corruption guarantee)**.** *Fix $\zeta > 0$. There is a polynomial P_ζ, whose degree and constants depend only on ζ, with the following property. Let D be any input distribution on the Boolean cube, let f be any Boolean target, and let g have the shallow representation in Lemma 23.1. Put $\rho = \Pr_D\{f(X) \neq g(X)\}$. Fix a learner depth $L \geq 2$, $0 < \varepsilon, \delta < 1$, and an integer patch budget $b \geq 4\log(2/\delta)$. For any integer sample size n satisfying both*

$$n \geq P_\zeta(d, L, C_b, R_b, \varepsilon^{-1}, \log(2/\delta)), \qquad \rho n \leq b/4,$$

the original Gaussian posterior, conditioned only on global clean output margin and on ordinary training signs, has Gibbs error at most ε with probability at least $1 - \delta$. The sample event is simultaneous for every actual admissible width $N \geq C_L(f)$ and every $0 < \sigma^2 \leq N^{-(1+\zeta)}$. No shallow representation or circuit-size bound for the exact target f is assumed.

Proof. The number of observed disagreements between f and g is binomial with mean $\rho n \leq b/4$. A Chernoff bound gives

$$\Pr\{\text{more than } b \text{ disagreements}\} \leq (e/4)^b \leq \delta/2.$$

On the complementary event, flip g at the distinct observed disagreement inputs. Lemma 23.1 supplies a globally unit-margin, sample-consistent shallow witness with the same deterministic parameter bounds C_b, R_b, independently of which disagreements occur.

Apply zero-padding, balanced cloning, and scalar transmission to this witness. With

$$M = \left\lceil \max\{4, C_b, 2R_b, dC_b(d+1)R_b^2\} \right\rceil, \qquad T = MC_bR_b^2,$$

one obtains unit-margin interpolants at every $N \geq M$, with energy $K_2 = 2 + 2T$ at depth two and $K_L = 2 + (L + 14)(2T)^{1/(L-1)}$ at depth $L \geq 3$. Their energy-product parameters are polynomial in d, C_b, R_b, uniformly over L.

The Gaussian-distance proof uses its witness only to lower-bound the raw probability of the posterior conditioning event. A globally margin-valid sample interpolant suffices for that step; its

labels away from the sample need not equal f. Therefore the same proof yields, on a sample event of probability at least $1 - \delta/2$, the desired risk bound whenever the disagreement budget is met. The uniform Rademacher and finite-grid sample events depend only on the displayed deterministic bounds, not on the locations or labels of the patched inputs. There is no union bound over patch sets in this argument.

For completeness, this also handles every smaller actual admissible width. The Gaussian-distance argument has a deterministic polynomial width cutoff $W \geq M$. For $C_L(f) \leq N < W$, a defining width-$C_L(f)$ realization of the true target has weights at most $C_L(f) \leq W$. Clone it to width N and double its readout. Its weights are at most $2W$, and its energy and Gaussian-box target-mass exponent are polynomial in d, L, W. The finite-grid proof therefore applies uniformly to this width range. If $C_L(f) \geq W$, that range is empty. In either case its sample bound has the stated polynomial form and does not require an upper bound on $C_L(f)$ as a separate parameter. Combining the two sample events proves the theorem. □

For example, choose $b = \lceil 4 \log(2/\delta) \rceil$ and take n to be the ceiling of the displayed polynomial bound. The permitted corruption level $b/(4n)$ is inverse polynomial in the base representation and the statistical parameters. This is stronger than the elementary comparison that requires observing no disagreement, whose failure probability is controlled only when $\rho n \lesssim \delta$. Here a nonzero number of contradictory training labels is explicitly interpolated. The two inequalities on n are essential: this argument does not imply that the guarantee persists for every larger sample size at a fixed positive corruption probability.

24 A limitation of neuronwise cloning, even at polynomial path weight

The previous representation limitation based on absolute path weight does not exhaust the obstruction to improving the intrinsic-depth exponent. Even a circuit with polynomial absolute path weight can have a quasipolynomial minimum product of layer Frobenius norms under *all* neuronwise cloning choices. The statement below concerns an exact representation-preserving method; it is not a lower bound on the minimum energy of the represented Boolean function, which may have a different and much cheaper realization.

Proposition 24.1 (Complete-tree cloning obstruction)**.** *Let $H \geq 1$, $d = 2^H$, and consider a complete binary computation tree with H hidden layers. Layer s contains $n_s = 2^{H-s}$ computational neurons. Each neuron has two computational incoming edges of absolute weight $a > 0$; the first layer takes its two inputs from distinct Boolean coordinates. The final scalar readout has absolute weight $\beta > 0$. Arbitrary additional bias-supply neurons and edges are permitted.*

Replace every computational neuron by any positive integer number of identical copies, divide its outgoing edges by the number of its copies, and fill unused positions with zero neurons to obtain common hidden width N. Write $E_1, \ldots, E_{H+1}$ for the weighted squared layer norms of the resulting network. Then

$$\prod_{s=1}^{H+1} E_s \geq d\beta^2 (2a^2)^H 2^{H(H-1)/2}.$$

Consequently its Gaussian energy satisfies

$$\kappa_N \geq (H+1) \left[d\beta^2 (2a^2)^H 2^{H(H-1)/2} \right]^{1/(H+1)}.$$

The computational input-to-output absolute path sum is merely $\beta(2a)^H$. For fixed $a, \beta > 0$, this is polynomial in d, whereas the lower bound on the product of layer energies is $\exp(\Omega((\log d)^2))$.

Proof. For a computational node v, let $r_v > 0$ denote its number of copies divided by N. Discard all non-computational edges when lower-bounding layer energies. The remaining contributions are

$$F_1 = 2da^2 \sum_{v \text{ at level } 1} r_v, \qquad F_s = a^2 \sum_{v \text{ at level } s} \sum_{u \text{ a child of } v} \frac{r_v}{r_u} \quad (2 \le s \le H), \qquad F_{H+1} = \frac{\beta^2}{r_{\text{root}}}.$$

Each F_s is a convex function of the log multiplicities $y_v = \log r_v$, being a sum of exponentials of affine functions. Average y over the finite automorphism group of the complete binary tree. The group preserves every F_s and acts transitively on each level, so Jensen's inequality decreases every F_s simultaneously and makes $r_v = r_s$ constant on level s. This averaging is used only for a lower bound and need not give integer clone counts. The width constraints also remain valid, by convexity of the exponential, although that fact is not needed for the lower bound.

At these averaged log multiplicities,

$$F_1 = 2da^2 n_1 r_1, \quad F_s = 2a^2 n_s \frac{r_s}{r_{s-1}}, \quad F_{H+1} = \frac{\beta^2}{r_H}.$$

Their product telescopes to $d\beta^2(2a^2)^H \prod_{s=1}^H n_s$. Since $E_s \ge F_s$ before averaging and every F_s decreases on averaging, this lower-bounds the original product of layer energies. Now $\prod_s n_s = 2^{H(H-1)/2}$, and arithmetic–geometric mean gives the energy bound. There are 2^H computational input-to-output paths, each with absolute weight βa^H, proving the last assertion. □

In particular, allowing nonuniform clone counts within a layer does not turn a polynomial absolute path bound into the polynomial energy-product bound needed by the unrestricted Gaussian-distance theorem. A proof of that implication would need a change of representation beyond exact cloning, or a different posterior argument.

25 What remains unresolved

The necessary condition concerns functions up to approximation, not a particular implementation. The depth-independent product normal form proves that ordinary Boolean description size is polynomial in intrinsic product complexity. The converse polynomial comparison is not known, and even such a comparison would retain the learner-depth qualification in our present sufficiency theorem. The exact-cloning obstruction above therefore does not supply the missing function-level lower bound. Conversely, the shallow and halfspace results do not identify shallow representation size with arbitrary deep circuit size. A joint polynomial theorem for every small deep target, or an explicit counterexample to it under this posterior, remains open.

The exponent $\gamma = 1$ lies outside the Gaussian-distance argument: its typical hidden Gaussian bulk has order-one operator norm, so the uniform subtraction estimate used here does not become small with width. This observation is a proof limitation, not a sample lower bound. Removing the original all-input output-margin conditioning is also outside the present statements.

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